The World Islamic Sciences and Education University

جامعة العلوم الاسلامية العالمية

Faculty of Information Technology

كلية تكنولوجيا المعلومات

<u>GRADUATION PROJECT</u>

Title

Ro5stak: Centralized Driver’s License Platform for Theoretical Preparation and Mentor Recommendation

Students

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Abstract

This project presents the design and development of a centralized digital platform aimed at simplifying and improving the process of obtaining a driver’s license in Jordan. The platform provides an integrated environment that combines structured theoretical learning, guided practical training, mentor discovery, and progress tracking within a single system.

The proposed solution addresses the current challenges faced by trainees, including the lack of clear guidance, limited access to structured educational resources, and inconsistency in practical training quality. Through the integration of learning modules, booking systems, communication tools, and optional payment services, the platform enables users to navigate the licensing process in a more organized and efficient manner.

Additionally, the system incorporates features such as simulated government API integration, multilingual support, and AI-assisted guidance to enhance user experience and accessibility. The overall objective is to improve learning outcomes, increase transparency, and promote safer driving practices among new drivers.

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Chapter 1: Introduction

1.1 Overview

Obtaining a driver's license in Jordan requires applicants to navigate multiple stages, including registration, theoretical preparation, medical evaluation, and practical driving training. While these steps are essential, the overall process is often experienced as fragmented and unclear, with limited centralized guidance or structured support for trainees.

In practice, applicants frequently rely on informal sources such as instructors, friends, or online materials to understand the process and prepare for examinations. This results in inconsistent access to information, varying learning experiences, and a lack of standardization in both theoretical and practical training.

To address these challenges, this project proposes the development of a centralized digital platform that streamlines the entire licensing journey. The platform is designed to provide a structured and user-centered experience by integrating step-by-step guidance, interactive theoretical learning modules, progress tracking, and a transparent system for discovering and engaging with driving instructors.

By consolidating these services into a single platform, the system aims to simplify the licensing process, enhance the quality and accessibility of learning resources, and promote a more consistent and effective training experience for all users.

1.2 Problem Statement

The process of obtaining a driver's license in Jordan lacks a structured and centralized system that effectively guides trainees through the required stages of theoretical preparation and practical training. As a result, applicants are often left to navigate the process independently, leading to confusion, inefficiency, and inconsistent learning outcomes.

To better understand these challenges, a survey was conducted targeting individuals who are either currently undergoing the licensing process or have obtained their license within the past three years. A total of 50 responses were collected, of which 40 met the target criteria and were included in the analysis. The findings highlight several critical gaps in the current system.

Over 75% of respondents reported that they did not fully understand the licensing procedure before starting, and 40% required at least one hour to comprehend the necessary steps, with a substantial portion taking over three hours. Furthermore, approximately 88% relied on external assistance—such as friends, family members, or instructors to understand the process, and nearly half indicated they would not have been able to proceed without such support. These findings reflect a lack of accessible, clear, and centralized guidance for applicants.

In addition to procedural confusion, the theoretical preparation process is largely unstructured and ineffective. More than half of respondents (52.5%) reported relying on memorizing question banks rather than understanding underlying concepts, while 50% experienced difficulty finding reliable study materials. Consequently, a majority of trainees (65%) reported only partial or minimal understanding of traffic laws after passing the theoretical exam, indicating that the current approach prioritizes short-term exam success over long-term knowledge retention and safe driving awareness.

Practical driving instruction also suffers from inconsistency and lack of transparency. Approximately 82.5% of respondents described their training as either only somewhat structured or completely unstructured, with sessions often lacking a clear progression plan. Additionally, 70% of trainees found their instructors through informal channels such as word of mouth, and more than half received little to no information about their instructor’s qualifications or teaching style prior to starting. This absence of standardization limits trainees’ ability to make informed decisions and contributes to variability in training quality.

Overall satisfaction levels further reflect these challenges, with a majority of respondents rating the clarity and organization of the licensing process poorly. At the same time, there is strong demand for improvement: over 85% of respondents indicated that a centralized digital platform would have at least somewhat improved their experience, with more than half stating it would have addressed significant gaps.

These findings demonstrate a clear need for a unified system that provides structured guidance, accessible learning resources, and transparent access to training services. Without such a system, trainees remain dependent on informal support networks, experience inconsistent training quality, and may lack a comprehensive understanding of safe driving practices. Addressing these gaps is essential to improving both the efficiency of the licensing process and the overall quality of driver education in Jordan. **This gap not only affects user experience, but also has broader implications for road safety and driver preparedness.**

1.3 Project Objectives

- **To** provide a centralized platform that guides users through the entire driver licensing process in a structured and user-friendly manner.
- **To** deliver structured theoretical and practical learning resources that enhance users’ understanding of traffic laws and driving skills.
- **To** facilitate efficient interaction between trainees and certified mentors through booking, communication, and performance tracking features.
- **To** improve user experience and learning outcomes by integrating progress tracking, personalized recommendations, and system-assisted guidance.

1.4 Research Strategy

This project adopts an Agile Scrum methodology, which focuses on iterative development, continuous feedback, and incremental delivery of system features. The choice of this approach is based on the dynamic nature of the project requirements and the need to develop a functional system that evolves through multiple development cycles rather than a single linear process.

Through this methodology, the project is divided into a series of time-boxed sprints, where each sprint delivers a functional increment of the system. Each sprint includes planning, design, implementation, testing, and review phases, ensuring continuous improvement and adaptation throughout the development lifecycle.

The Scrum framework enables the team to prioritize features using a product backlog, refine requirements progressively, and respond effectively to changes or feedback from stakeholders. Regular activities such as Sprint Planning, Daily Scrum meetings, Sprint Reviews, and Sprint Retrospectives ensure transparency, collaboration, and consistent progress.

This approach is particularly suitable for this project as it allows the development of a complex, user-centered platform in a flexible and controlled manner, while ensuring that each iteration delivers measurable value. By adopting Scrum, the project achieves both practical implementation and continuous evaluation, enhancing system quality, usability, and alignment with user needs.

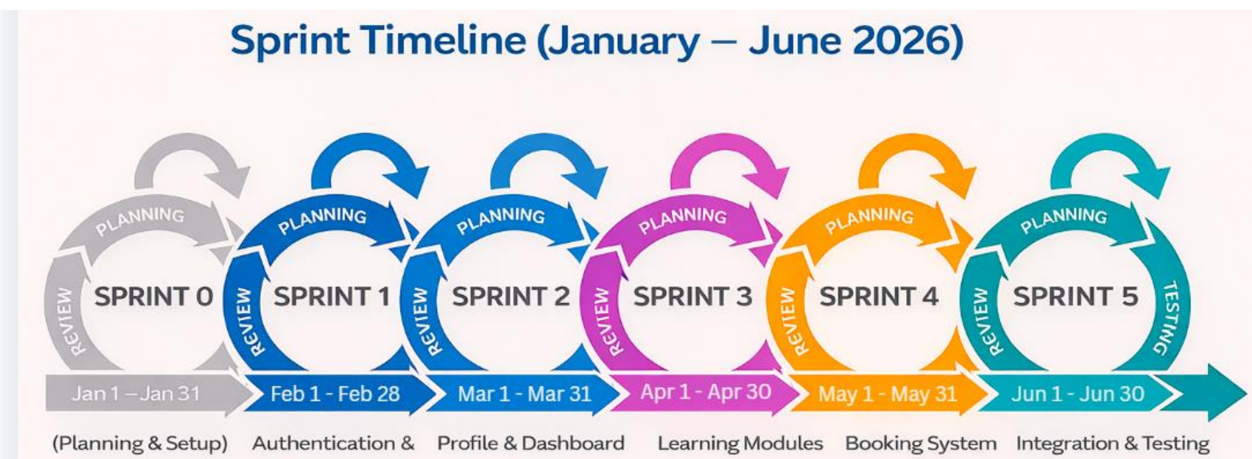
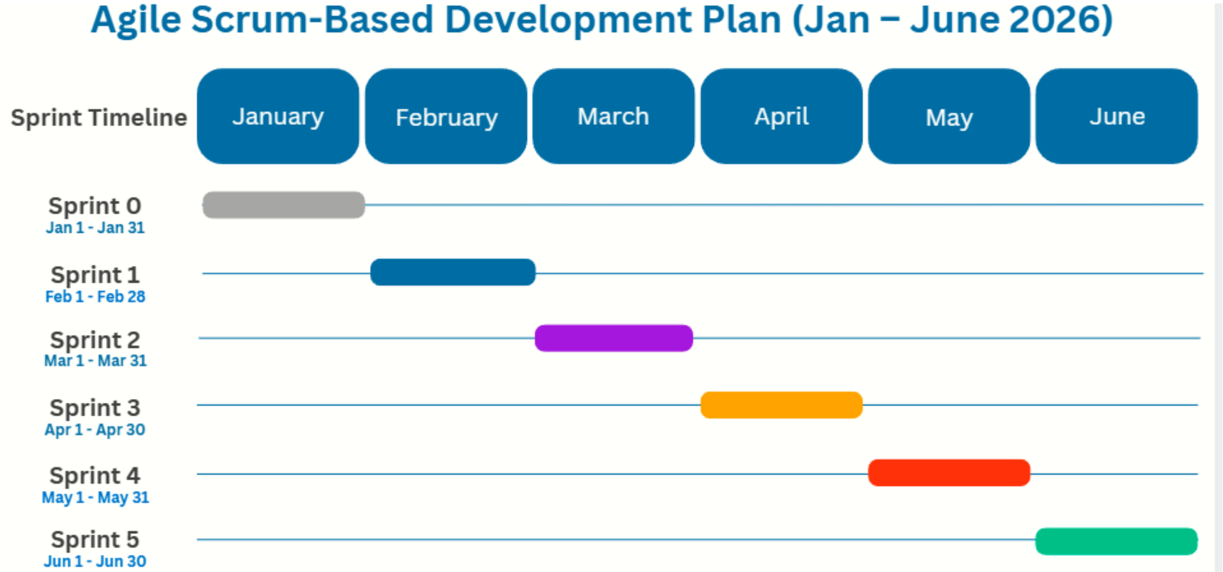


Figure 1.1: Scrum Framework

1.5 Gantt Chart

Table 1.1: Sprint Timeline and Key Deliverables (Gantt Chart)



1.6 Scope (Boundary)

The system focuses on providing a centralized digital platform that supports users throughout the driver licensing process, including:

- User registration and profile management
- Structured theoretical learning modules and quizzes
- Mentor discovery, filtering, and booking
- Appointment scheduling and session management
- Progress tracking and dashboards
- Messaging between trainees and mentors
- Notifications and reminders
- Simulated integration with government systems
- Optional payment and billing functionality

Out of Scope:

- Real-time integration with actual government systems (only simulated APIs are used)
- Physical driving test execution
- Direct management of driving training centers
- Hardware-based vehicle tracking systems

This defined scope ensures that *Ro5stak* remains focused, achievable, and aligned with project constraints, while laying a foundation for potential future enhancements.

1.7 Project Outline

- **Chapter 1**: Introduces the research problem, presents the project background, defines the objectives, and explains the research strategy and methodology used in the project.
- **Chapter 2**: Reviews related work and existing systems in the domain of driving license platforms and e-learning systems and highlights the need for the proposed solution.
- **Chapter 3**: Presents the system analysis, including feasibility study, functional and non-functional requirements, and an overview of system specifications.
- **Chapter 4**: Describes the system design, including database design (ERD), system architecture, and UML diagrams such as use case, sequence, and activity diagrams.
- **Chapter 5**: Explains the implementation phase, including technologies used, system development, and integration of different components, followed by testing and validation of system functionality.
- **Chapter 6**: Concludes the project by summarizing key results, discussing challenges faced during development, and providing recommendations for future improvements and extensions.

Chapter 2: Literature Review

2.1 Overview

This chapter presents a review of existing literature and related systems relevant to the development of Ro5stak, an online platform designed to guide individuals through every stage of obtaining a driving license, from discovering and booking a mentor to scheduling the practical exam. Section 2.2 examines a selection of related systems and platforms that share functional or conceptual similarities with Ro5stak, including government licensing portals, mentorship platforms, and AI-powered learning management systems. Section 2.3 provides a structured comparative analysis of these systems, evaluating them against key criteria relevant to the goals of Ro5stak. Section 2.4 summarizes the key findings of the review, identifies the gaps present in existing solutions, and explains how Ro5stak is positioned to address them.

2.2 Related Work

2.2.1 MOI UAE (Ministry of Interior – UAE)

The MOI UAE platform is an official government digital service provided by the Ministry of Interior of the United Arab Emirates. It was developed to allow citizens and residents to access a wide range of government services electronically, without needing to visit physical service centers. The application covers three main sectors: Traffic and Licensing, Civil Defense, and Police Headquarters, offering services such as driver licensing, vehicle licensing, traffic fines management, and appointment booking. In relation to Ro5stak, the MOI UAE platform is particularly relevant as it handles the official side of the driving license journey, including the ability to open a traffic file to attend theoretical and practical training courses in order to obtain a driving license. While the platform excels in connecting users to official government procedures, it does not provide any guidance, learning content, or mentor booking features to help users prepare for those procedures. There is no mentorship matching, no study modules, no quiz preparation, and no support system for trainees who need help navigating the license process step by step. Ro5stak is designed to fill this gap by serving as the preparation and guidance layer that sits alongside government portals like MOI UAE, helping users arrive at each official step fully informed and ready.

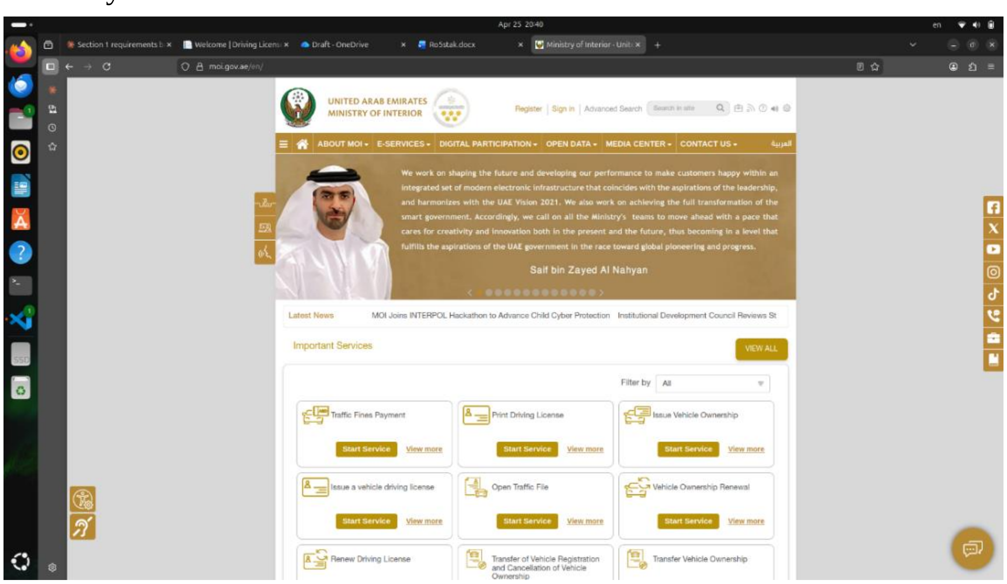


Figure 2.1: MOI UAE Platform Overview

2.2.2 Qooper Mentorship Platform

Qooper is an AI-powered mentorship and people development platform designed primarily for organizations, universities, and enterprises seeking to run structured mentoring programs at scale. The platform provides features for designing, running, and scaling mentoring programs with AI-assisted matching, training, guidance, and reporting, built for real organizational outcomes. Qooper uses a fully customizable automatic matching platform that includes auto-matching, admin matching, and self-matching options, pairing mentors with mentees based on skill gaps, goals, and preferred matching behavior. Additionally, mentors and mentees can schedule meetings, achieve goals, and complete training, earning badges and certification through Qooper's reward system. Despite these strengths, Qooper is built for professional and organizational contexts, meaning it is not tailored to domain-specific mentorship such as driving license training. It lacks any integration with government licensing systems, does not support practical exam booking, and provides no driving-specific learning content or quizzes. Its matching algorithm is designed around career and professional development goals, not the specific needs of a driving trainee seeking a certified mentor. Ro5stak addresses this gap by offering mentor matching and booking within a focused, domain-specific context directly tied to the driving license journey.

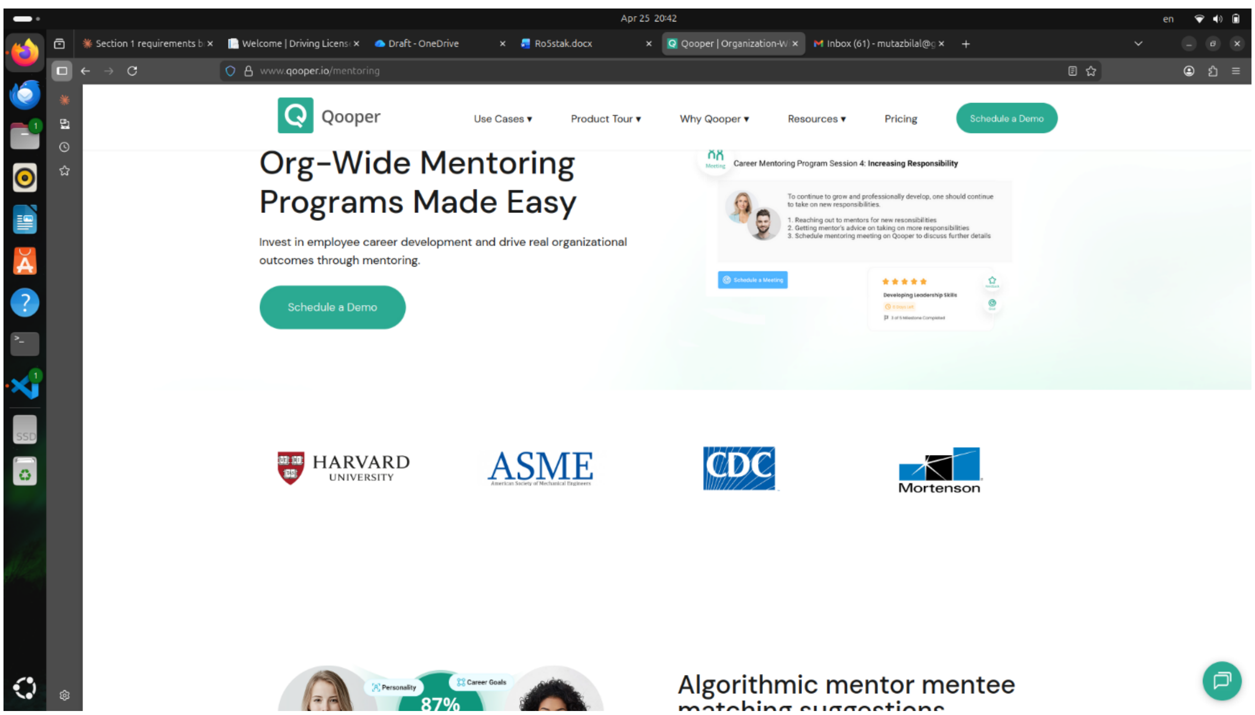


Figure 2.2: Qooper Mentorship Platform Overview

2.2.3 360Learning (AI-Powered LMS)

360Learning is a cloud-based Learning Management System (LMS) that combines collaborative and AI-powered learning features for organizations aiming to develop their workforce. With minimal input such as a title, topic, or existing document, the platform's AI can suggest structured course outlines, generate clear learning objectives, write course descriptions, and recommend personalized learning paths based on learners' roles, skills, and prior interactions. The platform also supports AI-powered authoring that converts existing documents into structured courses and generates quizzes, while L&D teams maintain oversight through internal comment threads and validation workflows. However, 360Learning is built as a general-purpose enterprise training tool and is not tailored to individual learners pursuing a specific real-world goal such as obtaining a driving license. It does not offer mentor booking, appointment scheduling with certified instructors, government API integration, or any pathway that guides a user from learning through to taking an official practical exam. The platform supports English, French, and German languages, with no Arabic language support, making it inaccessible to a large portion of the target audience for Ro5stak. Ro5stak builds on the concept of structured learning modules and quizzes but embeds them within a complete end-to-end journey specific to driving license acquisition.

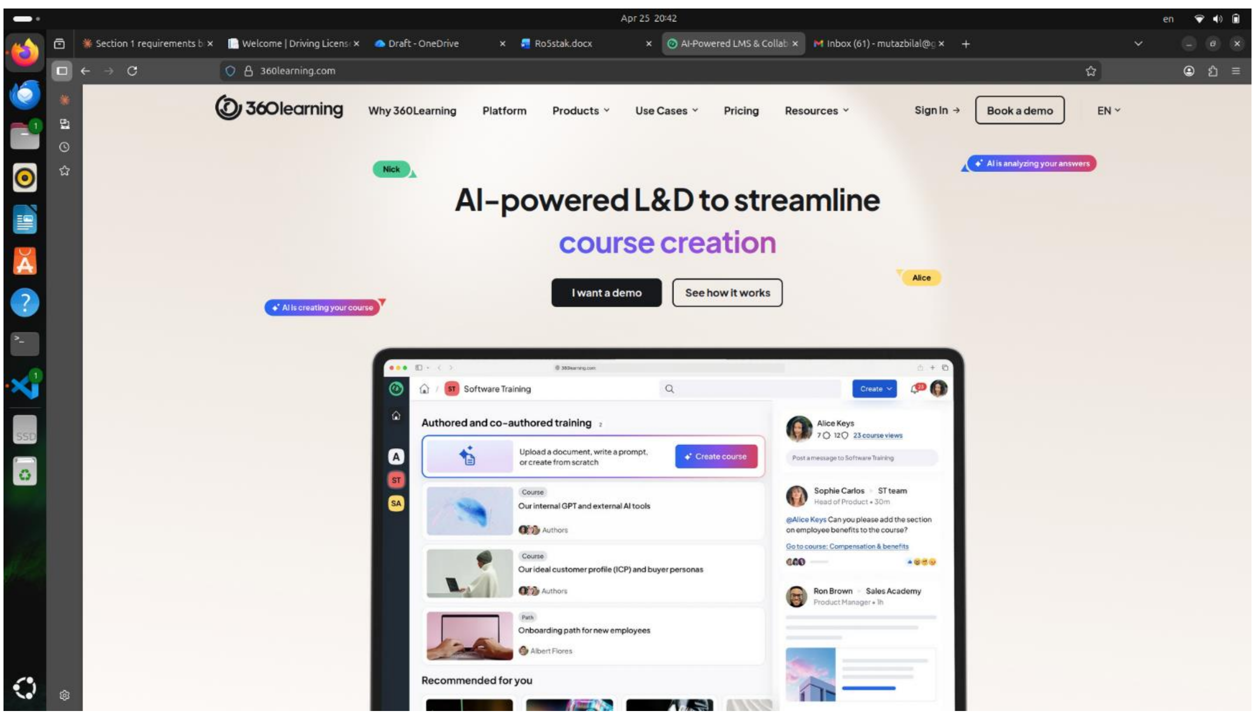


Figure 2.3: 360Learning Platform Overview

2.2.4 Mindgrasp AI

Mindgrasp AI is an AI-powered study tool designed to help students learn more efficiently by transforming uploaded content into interactive study materials. Users can upload any material and Mindgrasp builds an entire AI study session, including notes, flashcards, quizzes, a summary, and an AI tutor. Core features include AI Notes, an AI Tutor, AI Summarization, an AI Quiz Maker, and an AI Flashcard Maker, all designed to support faster and more effective learning. The platform supports over 20 languages and is accessible across multiple devices, making it broadly accessible. However, Mindgrasp is a content-agnostic study aid rather than a structured training platform. It does not offer mentor matching, session booking, progress tracking tied to a real licensing process, or integration with government services. It relies entirely on user-uploaded materials, meaning a trainee would need to source their own driving theory content before the tool can be of any use. There is no built-in curriculum for driving license preparation, no certified mentor network, and no connection to practical exam booking. Ro5stak differs by providing a fully curated, domain-specific learning environment where content, mentors, and official exam scheduling are integrated into a single platform.

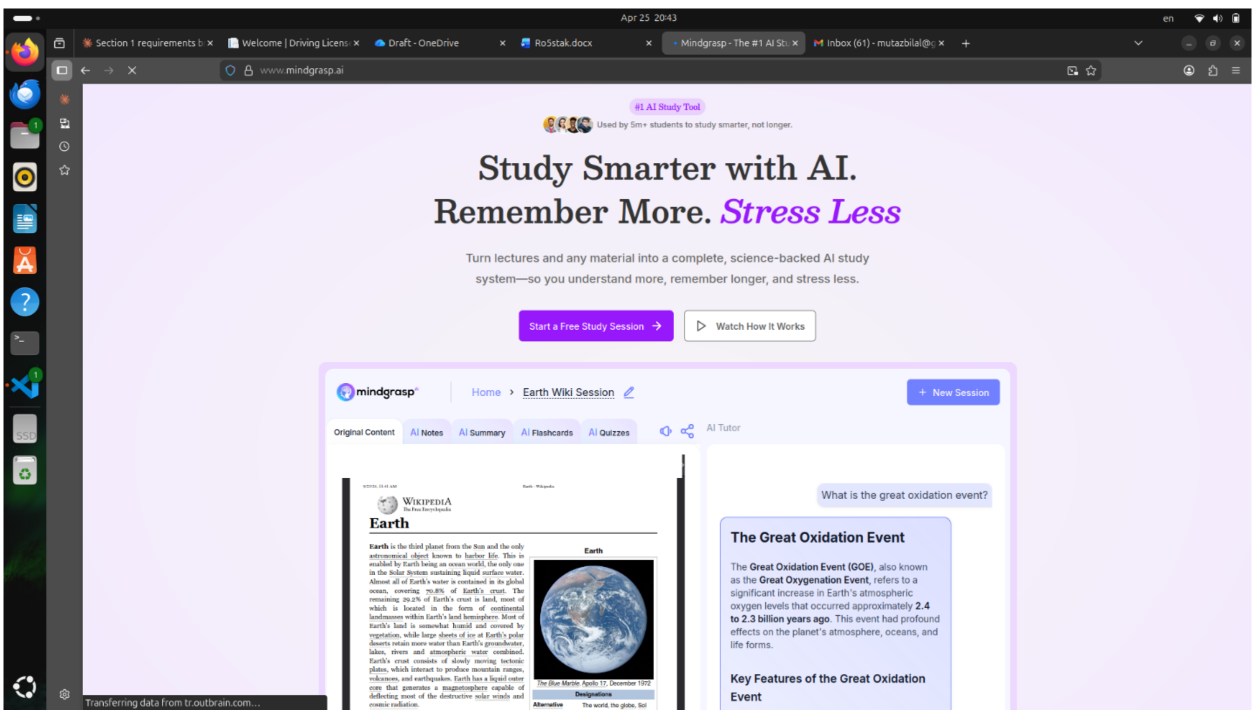


Figure 2.4: Mindgrasp AI Platform Overview

2.2.5 Edumentors

Edumentors is a UK-based online tutoring platform that connects students with tutors from top UK universities such as Oxford and Cambridge. The platform offers one-to-one tutoring sessions for children aged 4 to 19, covering a wide range of academic subjects, and includes an AI marking tool that provides instant feedback with personalized recommendations. Tutors are rigorously vetted through a multi-stage process that includes interviews and qualification verification, and the platform encourages trial sessions to help students find the right tutor match before committing to full sessions. Edumentors demonstrates a strong model for connecting learners with expert individuals in a one-to-one format, which is conceptually similar to Ro5stak's mentor booking feature. However, Edumentors is exclusively focused on academic school and university subjects and has no relevance to vocational or licensing-based training. It provides no driving-related content, no government service integration, no practical exam scheduling, and is geographically focused on the UK market with no Arabic language support. Ro5stak takes the core concept of personalized mentor-to-learner matching seen in platforms like Edumentors and applies it specifically to the driving license domain, extending it with exam booking, learning modules, and government integration.

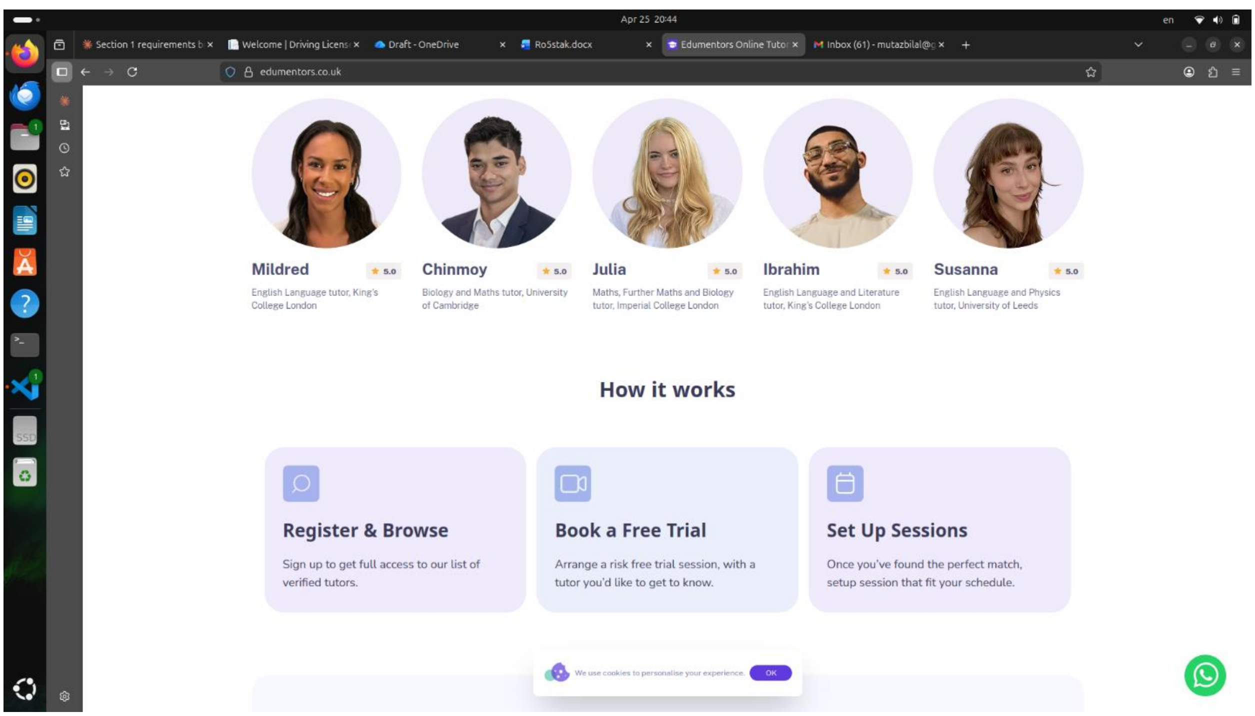


Figure 2.5: Edumentors Platform Overview

2.3 Comparison of Product Review Platforms

The following table presents a structured comparison of the five reviewed systems against nine key features relevant to the development of Ro5stak. The purpose of this comparison is to identify the functional gaps present in existing solutions and to highlight how Ro5stak addresses those gaps.

Table 2.1: Comparation between Ro5stak and other related works

Feature	MOI UAE	Qooper	360Learning	Mindgrasp AI	Edumentors	Ro5stak
Government API Integration	✓	✗	✗	✗	✗	✓
Mentor/Tutor Booking with Availability	✗	✓	✗	✗	✓	✓
Domain-Specific Content (Driving)	Partial	✗	✗	✗	✗	✓
Learning Modules with Quizzes	✗	Partial	✓	✓	✗	✓
AI Chatbot / Assistant	✗	✗	Partial	✓	Partial	✓
Exam Scheduling and Booking	✓	✗	✗	✗	✗	✓
Multi-Role Support (Trainee / Mentor / Admin)	Partial	✓	✓	✗	Partial	✓
Arabic Language Support	✓	✗	✗	Partial	✗	✓
Progress Tracking	✗	✓	✓	Partial	Partial	✓

2.3.1 What Existing Systems Do Well

The reviewed systems collectively demonstrate competence in isolated areas of the learning and mentorship landscape:

- **Government Integration & Exam Scheduling** MOI UAE stands out as the only platform with direct government integration and official exam scheduling, reflecting its purpose as an official state portal rather than a user-facing guidance tool.
- **Structured Mentorship & Learning Management** Qooper and 360Learning show strength in their respective domains, offering features such as AI-powered mentor matching, progress tracking, and quiz-based learning modules designed for organizational development.
- **AI-Powered Study Assistance** Mindgrasp AI contributes a strong AI-driven study model, allowing users to generate notes, flashcards, and quizzes from uploaded content, demonstrating the value of intelligent learning support tools.
- **One-to-One Tutor Booking** Edumentors presents a well-executed tutor discovery and booking experience, connecting learners with vetted tutors through a straightforward and accessible interface.

2.3.2 What Existing Systems Consistently Fail to Provide

Despite their individual strengths, the reviewed systems share critical gaps when evaluated against the specific requirements of a driving license guidance platform:

- **No Government API Integration in Non-Governmental Platforms** None of the non-governmental systems offer any connection to official licensing portals, so users must navigate all government procedures without any in-platform guidance.
- **No Unified End-to-End Driving Journey** No existing platform combines mentor booking, domain-specific driving content, and practical exam scheduling within a single unified experience, forcing users to rely on multiple disconnected tools.
- **Limited Arabic Language Support** Arabic language support is largely absent across all non-governmental platforms reviewed, creating a significant accessibility barrier for the intended user base of Ro5stak.
- **No Domain-Specific AI Assistant** The absence of an AI assistant tailored specifically to driving license queries is a gap that none of the current solutions adequately address.

2.3.3 How Ro5stak Addresses These Gaps

Ro5stak is specifically designed to address each of the identified gaps within a single cohesive platform:

- **Government API Connectivity** Ro5stak integrates directly with official government licensing systems, bridging the gap between user preparation and the formal procedures required to obtain a driving license.
- **Mentor Booking with Real-Time Availability** The platform allows trainees to browse, select, and book certified mentors based on live availability, providing a personalized and structured preparation experience.
- **Driving-Specific Learning Modules & Quizzes** Unlike general-purpose LMS platforms, Ro5stak delivers curated learning content and quizzes built specifically around driving theory and license preparation requirements.
- **Practical Exam Scheduling** Ro5stak enables users to schedule their official practical driving exam directly through the platform, completing the end-to-end journey without needing to visit external portals.
- **Dedicated AI Chatbot** A built-in AI assistant guides users through queries related to the licensing process, providing instant, context-aware support throughout their journey.
- **Arabic Language Support & Multi-Role Architecture** The platform is fully accessible in Arabic and supports a multi-role system serving trainees, mentors, and administrators, ensuring inclusivity and operational flexibility.

2.4 Summary

This chapter presented a review of five existing systems and platforms related to the domain of Ro5stak, covering government licensing portals, mentorship platforms, AI-powered learning tools, and online tutoring services. The systems reviewed were MOI UAE, Qooper Mentorship Platform, 360Learning, Mindgrasp AI, and Edumentors. Each system was examined in terms of its core purpose, key features, and most importantly, its limitations in relation to the goals of Ro5stak.

The comparative analysis revealed three critical gaps that are consistent across the majority of the reviewed systems. First, no existing non-governmental platform offers integration with official government licensing portals, leaving users without a bridge between their preparation journey and the official procedures they must complete. Second, none of the reviewed systems combine mentor booking, domain-specific driving content, and practical exam scheduling within a single unified platform, forcing users to rely on multiple disconnected tools to complete their licensing journey. Third, Arabic language support is largely absent from the non-governmental platforms reviewed, creating an accessibility barrier for a significant portion of the intended user base.

These findings confirm that a clear and well-defined gap exists in the market for a platform that guides individuals through every stage of obtaining a driving license in a cohesive, accessible, and intelligent manner. Ro5stak is positioned to address this gap by unifying government API integration, mentor booking, structured learning modules, AI assistance, practical exam scheduling, and multi-role support under one platform with full Arabic language support. The design decisions and system requirements that respond to these findings are presented and discussed in the following chapter, where the methodology adopted for the development of Ro5stak is outlined in detail.

Chapter 3: Methodology

3.1 Overview

This chapter presents the methodology adopted for the development of the proposed system. It outlines the systematic approach followed throughout the project lifecycle, starting from feasibility analysis and requirements engineering, to system design and implementation considerations.

The chapter begins with a feasibility study to evaluate the practicality of the system from technical, operational, and economic perspectives. This is followed by a detailed explanation of the requirements engineering process, including the methods used for data collection and the identification of system stakeholders.

Furthermore, the chapter describes the system design approach, including architectural decisions, database design considerations, and the selection of appropriate technologies. In addition, potential risks associated with the project are identified and analyzed.

Finally, for the intelligent component of the system, the methodology covers dataset preparation, model development, and evaluation processes, ensuring a comprehensive and structured approach to both software development and machine learning integration.

3.2 Feasibility Study

To assess the feasibility of the Driving License Training Platform prior to development, a structured feasibility study was conducted. The primary objective of this study was to estimate the system size, development effort, and overall complexity.

To achieve this, the Function Point Analysis (FPA) technique was adopted. FPA is a standardized method used to measure software size based on user-visible functionalities, independent of the technology used for implementation.

Based on the identified user stories, the system functionalities were categorized into five main components:

- External Inputs (**EIs**)
- External Outputs (**EOs**)
- External Inquiries (**EQs**)
- Internal Logical Files (**ILFs**)
- External Interface Files (**EIFs**)

Each component was classified based on its complexity level (Simple, Average, Complex), allowing for a structured estimation of development effort.

The following table presents the standard weights used in Function Point Analysis (FPA), categorized by complexity level. These weights are applied in the calculation column of the next table to estimate the function points for each functionality type in the system.

Table 3.1: Standard Weights Assigned to Function Point Components Based on Complexity

Component	Simple	Average	Complex
External Inputs (EIs)	3	4	6
External Outputs (EOs)	4	5	7
External Inquiries (EQs)	3	4	6
Internal Logical Files (ILFs)	7	10	15
External Interface Files(EIFs)	5	7	10

Function Point Estimation

The following sections present the detailed breakdown and calculation of the system's function points.

Table 3.2: Function Point Counts by Complexity

Component	Simple	Average	Complex
External Inputs (EIs)	8	10	6
External Outputs (EOs)	3	7	4
External Inquiries (EQs)	4	6	3
Internal Logical Files (ILFs)	5	6	3
External Interface Files(EIFs)	1	2	2

The following table presents a detailed breakdown of the function point calculations derived from the identified functionalities of the Driving License Training Platform. Each functionality was analyzed and assigned a complexity level (Simple, Average, or Complex) based on its processing logic, data handling, and interaction with other system components. The number of occurrences (Count) for each functionality was then multiplied by the corresponding standard weight for its complexity level. This calculation is illustrated in the (Count × Weight) column, while the resulting values are recorded as Function Points in the final column. These calculated values represent the relative development effort required for each functionality and collectively contribute to estimating the overall system size and development effort.

Table 3.3: Function Point Estimation Table

Functionality	Complexity	Calculation	Function Points
Inputs (External Inputs - EIs)			
User Registration / Mentor Registration	Complex	2 × 6	12
Login / Password Reset / Consent	Average	3 × 4	12
Profile Update / Language Preference	Simple	2 × 3	6
Booking Session / Cancel / Reschedule	Complex	3 × 6	18
Messaging / File Upload	Average	2 × 4	8
Quiz Submission / Mock Exam Start	Average	2 × 4	8
Exam Booking (Theory / Medical / Practical)	Complex	3 × 6	18
Admin Actions (Approve / Block Dates)	Average	2 × 4	8
Total Inputs			**90**
Outputs (External Outputs - EOs)			
Notifications (Booking / Exams / Messages)	Complex	3 × 7	21
Dashboard Display (Progress / Calendar)	Average	2 × 5	10
Certificate Generation	Complex	1 × 7	7
Admin Analytics & Reports	Average	2 × 5	10
AI Chatbot Responses	Average	2 × 5	10
Total Outputs			**58**
Inquiries (External Inquiries - EQs)			

Search Mentors / Filter System	Average	2 × 4	8
View Mentor Profile	Simple	1 × 3	3
View Schedule / Calendar	Average	2 × 4	8
View Trainee Progress	Complex	1 × 6	6
View Exam Results	Average	1 × 4	4
Total Inquiries			**29**
Internal Logical Files (ILFs)			
Users Table	Average	1 × 10	10
Mentors Table	Average	1 × 10	10
Bookings Table	Complex	1 × 15	15
Learning Modules & Quizzes	Complex	1 × 15	15
Messages & Files	Average	1 × 10	10
Reviews & Ratings	Average	1 × 10	10
Exam Records	Average	1 × 10	10
Total ILFs			**80**
External Interface Files (EIFs)			
Government Database	Complex	1 × 10	10
Email Service / Notifications	Average	1 × 7	7
AI Chatbot Integration	Complex	1 × 10	10
Total EIFs			**27**

The Unadjusted Function Points (UFP) are calculated as the sum of all function points from inputs, outputs, inquiries, ILFs, and EIFs:

UFP = 90 + 58 + 29 + 80 + 27 = **284**

Value Adjustment Factor (VAF)

After calculating the Unadjusted Function Points (UFP), it is necessary to refine this estimation by considering a set of General System Characteristics (GSCs) that influence the overall complexity of the system. These characteristics evaluate aspects such as performance requirements, data communication, user interaction, system distribution, and processing complexity.

Each characteristic is assigned a rating on a scale from 0 to 5, where 0 indicates no influence and 5 indicates a strong influence on the system. The sum of these ratings (ΣFi) is then used to calculate the Value Adjustment Factor (VAF), which adjusts the initial function point count to better reflect real-world development conditions.

The following table presents the evaluated ratings for each system characteristic in the Driving License Training Platform.

Table 3.4: Value Adjustment Factor (VAF)

#	General System Characteristic	Rating (0–5)
1	Data Communications	4
2	Distributed Data Processing	3
3	Performance Requirements	4
4	Heavily Used Configuration	3
5	Transaction Rate	4
6	On-line Data Entry	5
7	End-user Efficiency	5
8	On-line Update	5
9	Complex Processing	4
10	Reusability	3
11	Installation Ease	3
12	Operational Ease	4
13	Multiple Sites	2
14	Facilitate Change	4
Total ΣFi = 53		

The Value Adjustment Factor (VAF) is calculated using the standard formula:

VAF = 0.65 + (0.01 × ΣFi)

Substituting the calculated value:

VAF = 0.65 + (0.01 × 53) = 0.65 + 0.53 = **1.18**

The Adjusted Function Points (AFP) are then calculated by multiplying the Unadjusted Function Points (UFP) by the VAF:

AFP = UFP × VAF

AFP = 284 × 1.18 ≈ **335**

Assuming an average productivity rate of **25** Function Points per person-month, the estimated development effort is calculated as follows:

Effort (person-months) = AFP / Productivity Rate

Effort = 335 / 25 **≈ 14 person-months**

This estimation reflects the expected development effort required to implement Ro5stak Platform, taking into account both the functional size and system complexity.

3.2.1 Technical Feasibility

The proposed Platform is technically feasible due to the availability of mature and widely adopted technologies. The system is designed as a full-stack web application using ASP.NET Core for backend services, Angular for frontend development, and SQL Server for data management. These technologies provide high performance, scalability, and security, and are well-supported within modern development environments. The system architecture follows a layered and scalable design, allowing efficient handling of an initial user base of approximately 200–500 users, with the ability to scale beyond 1,000 users through load balancing and stateless API design.

From an infrastructure perspective, the system can operate efficiently on standard cloud resources, with optimized database queries and caching mechanisms ensuring response times under 300 ms. The integrated machine learning component requires moderate resources for training but supports real-time inference with minimal latency, making it suitable for live system integration. Additionally, the system incorporates secure authentication, role-based access control, and encrypted communication, ensuring compliance with standard security practices. Based on these factors, the system is considered technically feasible and within the capabilities of the development team.

3.2.2 Operational Feasibility

The system is operationally feasible as it aligns with the workflows and needs of its primary users, including trainees, mentors, and administrators. It supports essential operations such as registration, mentor discovery, session booking, communication, and progress tracking within a unified platform. The expected initial user base of 200–500 users, with moderate daily activity, can be efficiently handled by the proposed system without introducing operational complexity.

The platform is designed with usability in mind, allowing users to adapt quickly with minimal training due to its intuitive interface. It significantly reduces manual coordination efforts by automating scheduling, notifications, and communication, leading to an estimated reduction of 30–50% in administrative workload. Maintenance requirements are moderate and manageable by a small technical team, while operational risks are mitigated through flexible system design and user-friendly features. Therefore, the system is considered operationally feasible and suitable for real-world deployment.

3.2.3 Economic Feasibility

The economic feasibility of the system is supported by a balanced relationship between development cost, operational expenses, and expected benefits. The use of open-source technologies and existing development tools minimizes initial investment, with an estimated development cost ranging between $8,000 and $15,000 for a full-stack system with integrated AI capabilities. This cost is considered reasonable given the system's functionality and scalability potential.

Operational costs are relatively low, with estimated monthly expenses below $200, including hosting, database services, and machine learning operations. In return, the system offers significant benefits such as reducing administrative workload by up to 50%, improving efficiency, and enhancing user experience. Additionally, the platform has strong potential for future scalability and monetization, making it a valuable long-term investment. Based on this analysis, the system is economically feasible and capable of delivering substantial value relative to its cost.

Table 3.5: Realistic Budget Table for Ro5stak

#	Cost Category	Estimated Cost (JD)
1	Development Hardware	450
2	Dataset Acquisition	0
3	Software Development & AI Integration	400
4	Domain Registration	10
5	Frontend Hosting	25
6	Backend/API Hosting	30
7	Cloud GPU (Optional)	60
8	Miscellaneous (Testing & Tools)	25
Total Estimated Budget = 1000 JD		

3.3 Requirements Engineering

3.3.1 Requirements Collection Methods

The requirements of the proposed system were gathered using a combination of qualitative and quantitative approaches to ensure a comprehensive understanding of the problem domain.

Initially, a significant portion of the requirements was derived from real-life experiences, including the personal experiences of the development team as well as observations and feedback from relatives and acquaintances who have undergone the driver's licensing process in Jordan. These experiences provided valuable insights into common pain points such as lack of guidance, inconsistent training quality, and dependence on informal support.

To further validate these observations, a structured survey was conducted targeting individuals who are either currently undergoing the licensing process or have completed it within the past three years. A total of 50 responses were collected, with 40 valid responses included in the analysis. The results revealed that over 75% of participants lacked a clear understanding of the licensing process prior to starting, while approximately 88% relied on external assistance to navigate the process. Additionally, more than 50% of respondents reported difficulties in accessing reliable learning materials, and 82.5% described practical training as unstructured.

These findings strongly support the need for a centralized and structured platform, and they directly influenced the system requirements, particularly in areas such as guided workflows, learning modules, mentor selection, and communication features.

Future work may include conducting interviews with professional driving instructors and licensing authorities to further refine system requirements and validate assumptions.

3.3.2 Stakeholders Identification

The proposed system involves multiple stakeholders, each with distinct roles, expectations, and interactions with the platform.

The primary stakeholders are trainees, who represent the main users of the system. Their goal is to successfully obtain a driver's license through a structured, guided, and efficient process. They interact with the system by accessing learning materials, booking sessions, taking quizzes, and communicating with mentors.

The second key stakeholder group is mentors (driving instructors). Mentors use the system to manage their availability, conduct training sessions, monitor trainee progress, and provide feedback. They are also responsible for maintaining professional standards and ensuring effective knowledge transfer.

System administrators represent another critical stakeholder group. They are responsible for managing users, verifying mentor credentials, monitoring system activity, and ensuring smooth platform operation. Their role is essential in maintaining system integrity and enforcing platform rules.

In addition, government-related entities are considered indirect stakeholders. Although the system integrates with government services through simulated APIs, it must align with official licensing procedures and requirements.

Finally, system developers and maintainers are internal stakeholders responsible for maintaining, updating, and scaling the platform over time.

3.3.3 Functional Requirements

The functional requirements define the core functionalities and services that the system must provide to its users. These requirements describe how the system should behave and outline the interactions between different user roles, including trainees, mentors, and administrators. They are derived from the problem analysis, user needs, and the results of the conducted survey, ensuring that the system effectively addresses the identified gaps in the current driver's licensing process.

The following list presents the detailed functional requirements of the proposed system:

1. **User Registration**
   - 1.1 The system shall allow users to create accounts using National ID and password.
   - 1.2 Users shall first select account type (Trainee or Mentor), and the system shall present a registration form tailored to the selected type.

1.3 Users shall be able to register by providing their National ID, which is verified through a simulated government API (which is included in the system database).
1.4 During registration, the system shall automatically retrieve user personal details from the government database.
1.5 Users shall only be allowed to edit address after autofill. All other government-retrieved fields (name, DOB, gender, National ID) shall be locked.
1.6 The system shall allow users to select one of three license types:

- Motorcycle
- Private Automatic Car
- Private Manual Car

1.7 Users shall be able to select their license type during registration or later in their profile.
1.8 After registration, users shall access a dashboard with progress for a license they are currently in the process of taking.
1.9 Mentors shall be able to register with a separate authentication process using their professional credentials.
1.10 The system shall implement roles each with specific validity:

- Trainee
- Mentor
- Admin

1.11 During registration, users shall explicitly consent to government-data retrieval; consent record must be stored with timestamp.
1.12 Mentors shall upload scanned certification documents and licenses; system shall store documents and mark mentor ‘pending’ until admin approves.
1.13 Terms & Privacy Page + Data Retention: System shall present Terms of Use and Privacy Policy; define data retention periods for user data.

2. User Login:

2.1 Users shall be able to log in using their credentials and stay authenticated during their session.
2.2 Users shall be able to log in using their National ID and password.
2.3 Users shall be able to reset their password using the “Forgot Password?” button, by sending a link to the user's email.
2.4 After login, users shall access a dashboard with progress for their license type.

3. Profile Management:

3.1 Users shall be able to view their personal information, and update contact details, Email and Address.
3.2 Users should be able to update only their personal details that don’t conflict with the governmental details from the government database.
3.3 Mentors shall be able to set their weekly availability (days and times).

4. **User Dashboard:**

4.1 Users shall be able to track their progress for each step: theory test, medical exam, practical test.

4.2 Users shall be able to view a calendar with all upcoming appointments, deadlines, and exam schedules.

4.3 The system should save user progress when the user changes his license type to another type and creates a new dashboard for the new license type.

4.4 Multi-language UI: Users shall be able to use the platform in Arabic (default) and English.

5. **Mentor Dashboard:**

5.1 Mentors shall have Personal dashboards showing scheduling settings, calendars.

5.2 Mentors shall be able to view user progress and quiz results.

5.3 Mentors shall receive notifications when users submit quizzes or complete sessions.

5.4 Mentors shall be able to view their calendar with scheduled sessions and availability slots.

5.5 Mentors shall be able to confirm or reschedule appointments with users

5.6 Mentors shall be able to Mark scheduled appointments as done.

5.7 mentors shall send a certificate of completion once all sessions are done by the user.

6. **Admin Portal:**

6.1 Admins shall be able to manage users, mentors, content, bookings, and view system logs

6.2 Analytics: Admins shall have usage analytics (users list, bookings list and mentors list).

6.3 Admins should be able to view pending applications and approve or decline mentor applications.

6.4 Certification / Badge for Mentors: System shall display badges for mentors that passed verification.

6.5 The system shall track mentor performance analytics including:

- Average rating
- Completion rate
- Average time to license completion per trainee

6.6 The system shall track drop rates for each user.

7. **Learning Modules:**

7.1 The system shall provide video and text-based educational content covering traffic rules, signals, and laws specific to their chosen license type.

7.2 Users shall undergo quizzes after each learning module to test their understanding; quizzes need to be passed to move to the next lesson.

7.3 Progress Checkpoints / Milestones: System shall create milestone tasks (e.g., complete module 1, pass quiz 1) and mark them complete to show progress.

7.4 A mock exam is held after all the theoretical modules are complete.
7.5 Additional Learning modules are provided for the practical examination.
7.6 Students shall get recommendations on specific learning modules/ classes based on mentor feedback.
7.7 Completion (not passing) of the mock exam shall be required before a trainee can book a theory test.

8. Mentor Booking:

8.1 Users shall be able to browse a list of certified mentors, check availability, and book sessions.
8.2 A mentor that didn't pass verification shall not be visible to trainees.
8.3 The system shall provide a search function to find mentors by (city/area), rating, price, and vehicle type.
8.4 the system shall provide mentor performance analytics (average rating, and completion rate)
8.5 Mentors shall have a verification status (pending, approved, rejected), and only approved mentors shall be visible to trainees.

9. Appointment Scheduling:

9.1 Users shall be able to book appointments for:

- Mentor sessions
- Theory tests
- Medical exams
- Practical tests

Subject to the following constraints:

- o Theory test requires completion of all theoretical modules and mock exam
- o Medical exam requires passing the theory test
- o Practical test requires passing the medical exam and completing practical modules

9.2 The system shall block double-booking: if mentor already has booking at that slot, new booking is denied, and user offered next available slot.
9.3 Users shall be able to cancel or reschedule sessions up to 24 hours in advance.
9.4 The system shall allow admin to block certain days such as national holidays. Admin-blocked dates shall prevent new bookings only. Existing bookings shall NOT be automatically cancelled.
9.5 the system shall manage booking lifecycle states including:

- Pending
- Confirmed
- Completed
- Cancelled / Expired

9.6 The system shall automatically update booking statuses:

- Sessions become completed after end time
- Expired bookings are marked accordingly

10. In-App Messaging:

10.1 Users and mentors shall be able to message each other if a booking exists between them about sessions (with message history tied to booking)

10.2 Session Materials Upload: Mentors shall be able to upload session-specific attachments (PDFs, images) to user's session record.

11. Feedback System:

11.1 Mentors shall be able to upload notes and personalized feedback for each user's session.

11.2 Users shall be able to leave ratings and reviews for mentors after each session.

12. AI features:

12.1 The system shall offer an AI chatbot to answer FAQs and guide users through the platform.

12.2 The system shall recommend learning modules to trainees based on mentor feedback and quiz performance.

13. Integration with Government Services: The system shall integrate with a government database using simulations to synchronize exam appointments and results.

14. Notification system:

14.1 The system shall send notifications (in-app, email, or SMS) for:

- Booking confirmations
- Booking cancellations or rescheduling
- Upcoming appointments and exams
- Quiz submissions
- Session material uploads
- Mentor feedback submission
- Government exam updates (date changes/cancellations)

15. Validation & Data Integrity

15.1 The system shall enforce server-side validation on all inputs.

15.2 Invalid inputs shall return clear error messages.

15.3 Database constraints shall prevent invalid or inconsistent records.

16. Audit & Logging

16.1 The system shall log all critical admin actions including:

16.2 User/mentor deactivation

16.3 Mentor approval/rejection

16.4 Booking modifications and date changes

3.3.3 Non-Functional Requirements

The non-functional requirements define the quality attributes, constraints, and performance expectations of the system. These requirements ensure that the system operates efficiently, securely, and reliably under different conditions.

1. **Performance Requirements**
   - 1.1 The system shall respond to user requests within 2–3 seconds under normal load conditions.
   - 1.2 Critical API endpoints (e.g., login, booking, dashboard loading) shall have a response time of less than 300 milliseconds.
   - 1.3 The system shall support at least 50 concurrent users in the initial deployment phase.
   - 1.4 The system shall handle at least 200–500 registered users without performance degradation.
2. **Scalability Requirements**
   - 2.1 The system shall support horizontal scaling to accommodate future growth (up to 1000+ users).
   - 2.2 The backend shall be designed as stateless APIs to allow load balancing.
   - 2.3 The database shall support incremental growth of 10–20% per month.
3. **Security Requirements**
   - 3.1 The system shall use secure authentication mechanisms (e.g., JWT-based authentication).
   - 3.2 The system shall enforce role-based access control (RBAC) for Trainee, Mentor, and Admin roles.
   - 3.3 All data transmission shall be encrypted using HTTPS.
   - 3.4 Sensitive data (e.g., National ID, personal data) shall be securely stored and protected.
   - 3.5 The system shall log critical actions for auditing and monitoring purposes.
4. **Usability Requirements**
   - 4.1 The system shall provide an intuitive and user-friendly interface.
   - 4.2 New users shall be able to understand basic system functionality within 10–15 minutes.
   - 4.3 The system shall support multi-language functionality (Arabic as default, English as secondary).
   - 4.4 The system shall maintain consistent UI/UX across all pages and workflows.
5. **Reliability & Availability Requirements**
   - 5.1 The system shall achieve at least 95–99% uptime.
   - 5.2 The system shall handle failures gracefully without data loss.
   - 5.3 The system shall ensure data consistency during concurrent operations.
   - 5.4 Backup mechanisms shall be implemented to prevent data loss.

6. **Maintainability Requirements**
    6.1 The system shall follow a modular architecture (e.g., layered or clean architecture).
    6.2 The codebase shall follow SOLID principles and best practices.
    6.3 The system shall support easy debugging, updates, and feature extensions.
    6.4 Logging mechanisms shall be implemented to support issue tracking and debugging.
7. **Compatibility Requirements**
    7.1 The system shall be compatible with modern web browsers (Chrome, Edge, Firefox).
    7.2 The system shall support both desktop and mobile devices.
    7.3 The system shall maintain consistent functionality across different screen sizes.
8. **AI System Requirements**
    8.1 The AI chatbot shall respond within <1 second under normal conditions.
    8.2 Recommendation results shall be generated based on user performance and feedback data.
    8.3 Model inference shall not significantly impact system performance.

3.4 System Design Approach

3.4.1 Architecture Design

Ro5stak is developed as a web-based application following the Model-View-Controller (MVC) architectural pattern. This pattern was selected due to its clear separation of concerns, making the system easier to develop, maintain, and scale. In the MVC pattern, the Model handles all data logic and database interactions, the View is responsible for what the user sees and interacts with, and the Controller acts as the bridge between the two, processing user requests and returning the appropriate responses.

The system is composed of five core modules, each serving a distinct role within the platform:

- **Trainee Dashboard** The trainee dashboard serves as the primary interface for users pursuing their driving license. It displays the trainee's current stage in the licensing journey, whether that is the medical examination, theoretical training, or practical test. The dashboard also shows upcoming mentor sessions and scheduled exams, giving the trainee a clear and organized view of their progress at all times.

- **Mentor Dashboard** The mentor dashboard provides certified driving mentors with a dedicated space to manage their activity on the platform. When a trainee books a session with a mentor, the booking appears directly on the mentor's dashboard, allowing them to view upcoming sessions, manage their availability, and track their interactions with trainees.

- **Admin Panel** The admin panel gives platform administrators full oversight and control over the system. Through this panel, administrators can review and accept or reject mentor applications, ensuring that only qualified individuals are listed on the platform. Administrators can also block users who violate platform policies and manage exam booking availability by blocking specific dates when official exams are not available.

- **Booking System** The booking system acts as the connection layer between trainees and mentors. It allows trainees to browse available mentors, view their availability, and confirm session bookings. Once a booking is made, it is reflected on both the trainee's dashboard and the mentor's dashboard, ensuring both parties are informed in real time.
- **Learning Module** The learning module provides trainees with structured, domain-specific educational content tailored to driving license preparation. It includes lesson materials and quizzes designed to help trainees progress through the theoretical requirements of the licensing process. The module is integrated with the trainee dashboard so that progress is tracked and displayed as part of the overall licensing journey.

The interaction between these components follows the MVC flow. When a user performs an action, such as booking a mentor or completing a quiz, the request is sent to the Controller, which processes the logic and communicates with the Model to retrieve or update data. The result is then passed back to the View, which renders the updated interface for the user. This architecture ensures that Ro5stak remains modular, organized, and capable of supporting future enhancements without disrupting existing functionality.

3.4.2 Technologies Used

The development of Ro5stak involved a carefully selected set of technologies chosen to ensure a reliable, maintainable, and user-friendly web application. The following outlines the core technologies adopted across the frontend, backend, and database layers of the system.

- **Frontend — CSS** Plain CSS was used to design and build the user interface of Ro5stak. Rather than relying on a pre-built framework, custom CSS was written to give the platform a fully tailored visual identity that matches the specific design requirements of each module. This approach provided complete control over the styling of all interface elements including layouts, navigation, forms, dashboards, and interactive components, ensuring a consistent and purpose-built user experience across the entire platform.
- **Backend — ASP.NET** ASP.NET is a server-side web development framework developed by Microsoft, used to handle all business logic and server-side operations within Ro5stak. It was chosen for its robustness, strong performance, and seamless compatibility with SQL Server. ASP.NET supports the MVCS architectural pattern adopted by the system, allowing the codebase to be organized into clear layers of Models, Views, Controllers, and Services, each with a distinct responsibility within the application.

- **Architecture Pattern — MVCS (Model-View-Controller-Service)** The MVCS pattern extends the traditional MVC architecture by introducing a dedicated Service layer that handles business logic separately from the Controller. In Ro5stak, Controllers are responsible for receiving and directing user requests, while Services handle the processing and application logic before interacting with the data layer. This separation makes the codebase cleaner, easier to test, and more maintainable as the system grows.

- **Database — Microsoft SQL Server** Microsoft SQL Server was selected as the relational database management system for Ro5stak. It stores all platform data including user accounts, mentor profiles, booking records, learning module progress, and exam scheduling information. SQL Server was chosen for its strong integration with ASP.NET, its reliability in handling structured relational data, and its powerful querying capabilities through T-SQL. Its compatibility with the overall Microsoft technology stack used in the project made it the most suitable choice for the system.

3.5 Risk Analysis

Every software system faces potential risks that may impact its development, deployment, or long-term success. The following section identifies the key risks associated with Ro5setak, assesses their likelihood and impact, and outlines the mitigation strategies adopted to address each one.

- **Government API Integration Failure** One of the most critical dependencies in Ro5stak is its integration with official government licensing systems. If the government API is unavailable, poorly documented, or restricted, core features such as exam scheduling and license status tracking may not function as intended. To mitigate this risk, the system is designed with a modular integration layer that can be updated or replaced if the API changes, and alternative manual workflows are considered as a fallback.

- **Low User Adoption** As a new platform in a space where users may be unfamiliar with digital tools for license preparation, there is a risk that both trainees and mentors may be slow to adopt Ro5setak. This is mitigated through an intuitive user interface design, Arabic language support, and a simple onboarding process that minimizes friction for first-time users.

- **Mentor Availability and Quality** The value of Ro5stak depends heavily on having a sufficient number of qualified and active mentors on the platform. If mentor supply is low or inconsistent, trainees will not be able to book sessions effectively. This risk is addressed through the admin-controlled mentor application and approval process, which ensures quality, combined with a mentor management system that encourages active participation.

- **Data Privacy and Security** Ro5stak handles sensitive personal data including user profiles, booking records, and licensing progress. Any breach of this data could have serious legal and reputational consequences. This risk is mitigated by implementing secure authentication mechanisms, encrypting sensitive data, and following established data protection best practices throughout the development process.

- **System Scalability** As the number of users on the platform grows, the system must be able to handle increasing loads without performance degradation. Poor scalability could lead to slow response times or system crashes during peak usage. The MVC architecture adopted for Ro5stak supports modularity and separation of concerns, making it easier to scale individual components independently as demand increases.
- **Exam Date Management Conflicts** Since administrators manually control the availability of exam booking dates, there is a risk of scheduling conflicts or miscommunication between the platform and official exam centers. This is mitigated by giving administrators clear controls within the admin panel to block and manage dates in advance, reducing the likelihood of trainees booking on unavailable dates.
- **Internet Connectivity Dependency** As a web-based platform, Ro5stak requires a stable internet connection to function. Users in areas with poor connectivity may experience disruptions when accessing learning modules or booking sessions. While this cannot be fully eliminated, the system is designed to load efficiently and minimize unnecessary data usage to support users with slower connections.

3.6 Dataset

3.6.1 Data Source

3.6.2 Data Preprocessing

3.7 Model Development

3.7.1 Model Selection

3.7.2 Model Architecture

3.8 Model Development

3.8.1 Training Process

3.8.2 Evaluation Metrics

3.8.3 Results

Chapter 4: Design Models

4.1 Overview

This chapter presents the design models of the proposed Ro5stak platform. The purpose of this chapter is to illustrate the overall structure, behavior, and data organization of the system using a set of software engineering modeling techniques. These models help visualize how different system components interact with each other and provide a clearer understanding of the platform's architecture and workflows.

The chapter includes several diagrams and design representations, such as the Context Diagram, Data Flow Diagrams (DFDs), Use Case Diagram, Sequence Diagrams, Activity Diagrams, and Entity Relationship Diagram (ERD). Together, these models describe the system from different perspectives, including user interactions, process flow, data movement, and database structure.

4.2 Context Diagram

The Context Diagram (Level 0) represents the highest-level view of the Ro5stak platform. It illustrates the system as a single central process and shows how it interacts with external entities such as trainees, mentors, administrators, the simulated government API, the notification service, and the AI assistant service.

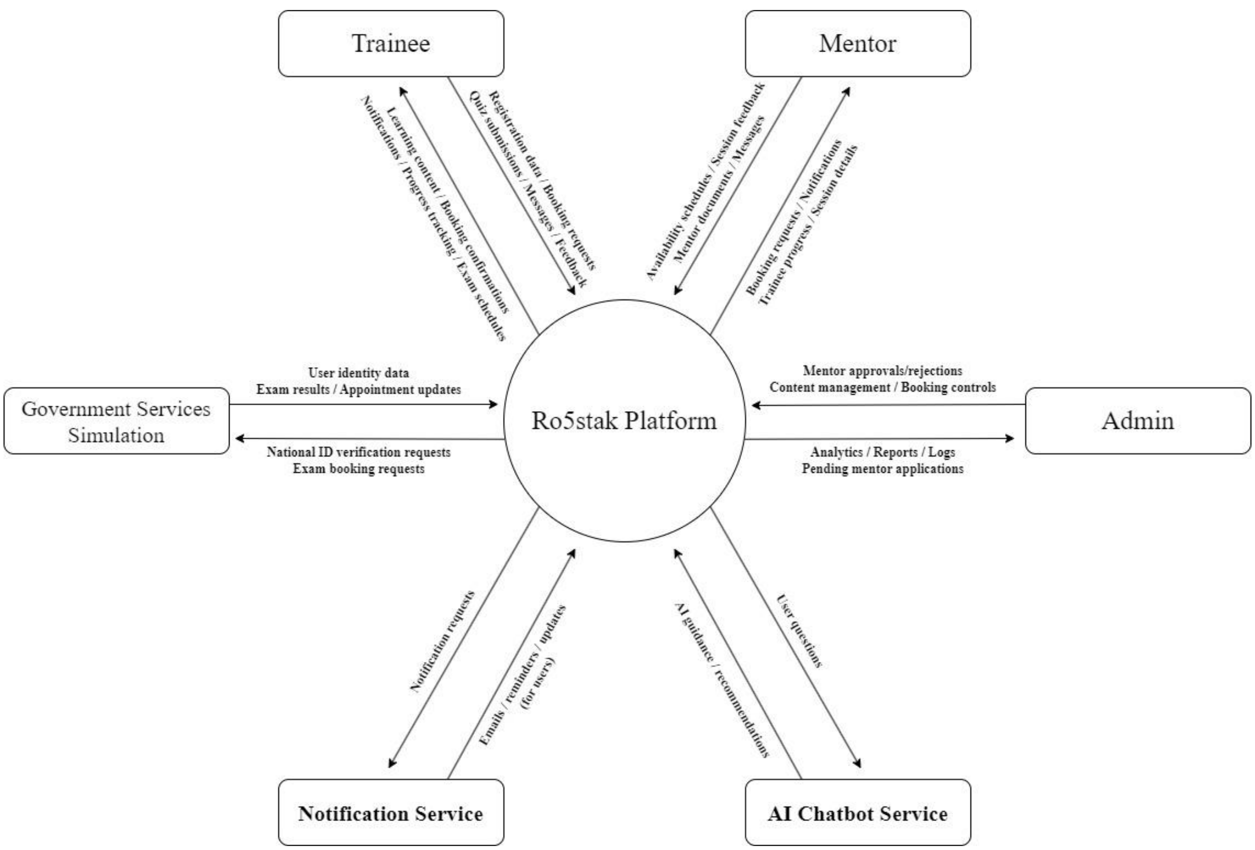


Figure 4.1: Context Diagram – Level 0

The purpose of this diagram is to provide a general understanding of the boundaries of the system and the flow of information between the platform and external actors. It highlights the main inputs and outputs exchanged without going into internal processing details.

4.3 Data flow Diagram

The Data Flow Diagram (DFD) Level 1 provides a more detailed representation of the internal processes within the Ro5stak platform. Due to the complexity and size of the system, the DFD was divided into multiple major processes, where each process focuses on a specific functional area of the platform.

The diagrams were divided into the following core processes:

Process 1.0 – User Management: Handles registration, login, profile management, and identity verification through the government services.

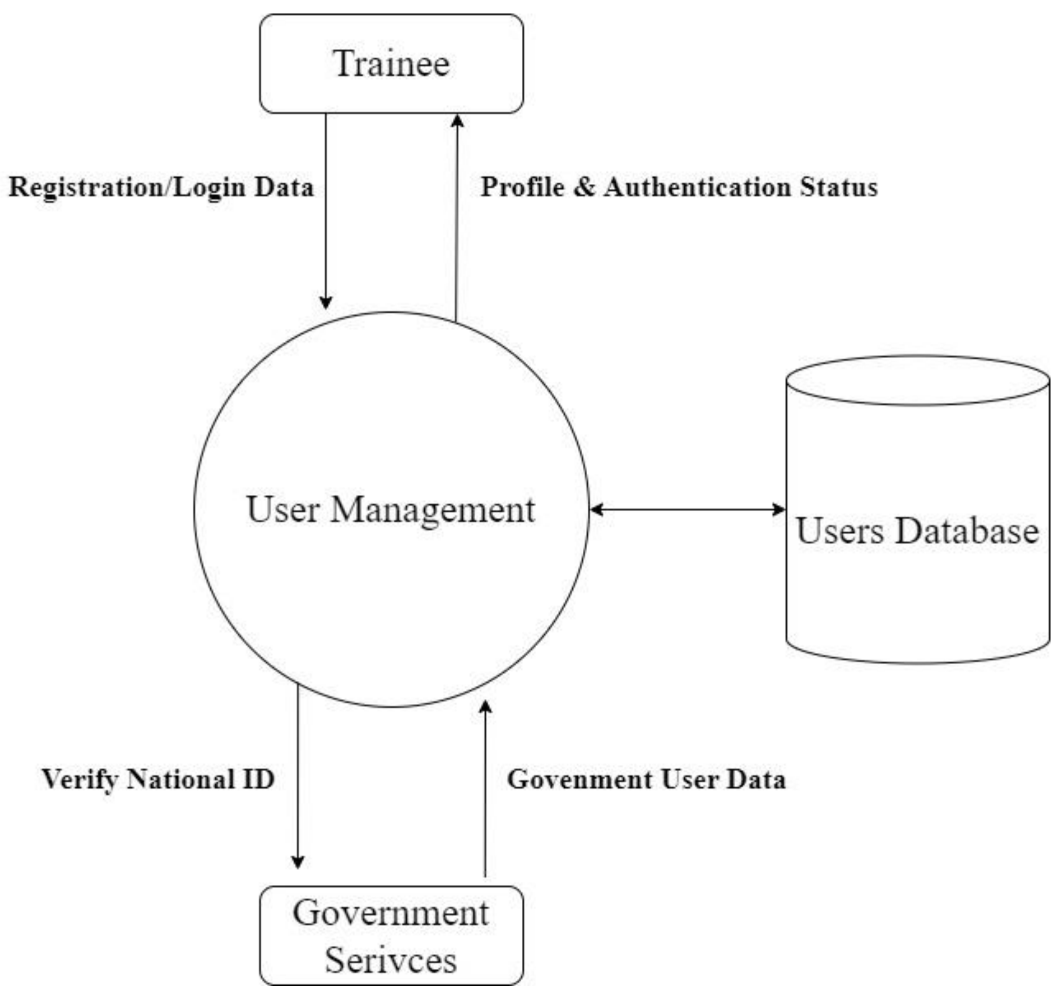


Figure 4.2: Data Flow Diagram – Process 1.0 User Management

Process 2.0 – Learning Management: Manages learning modules, quizzes, progress tracking, and AI-assisted recommendations.

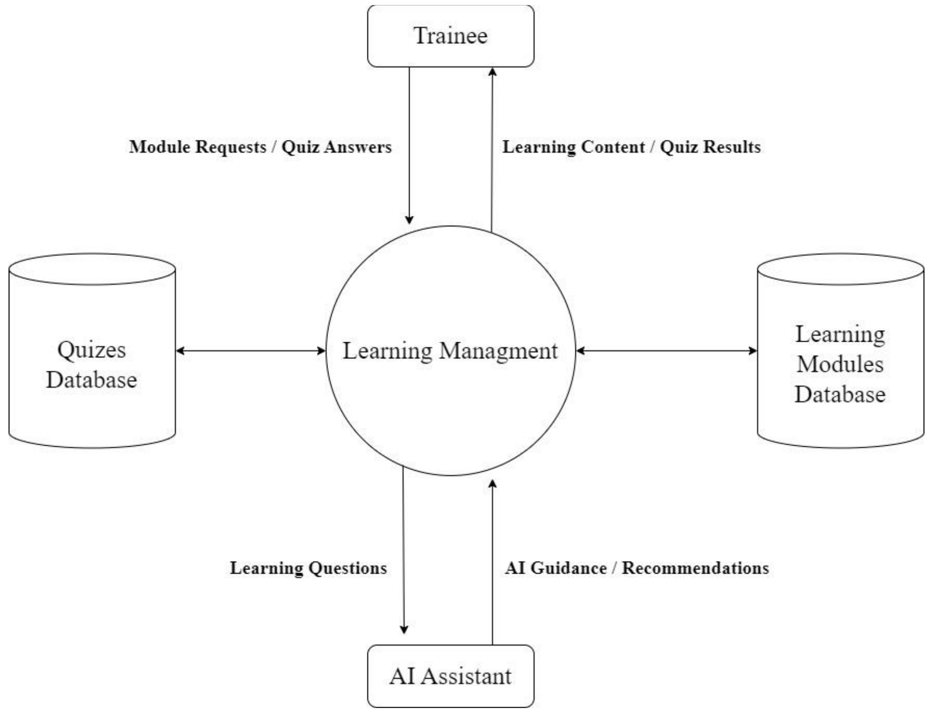


Figure 4.3: Data Flow Diagram – Process 2.0 Learning Management

Process 3.0 – Mentor Booking: Handles mentor discovery, availability management, booking sessions, and scheduling operations.

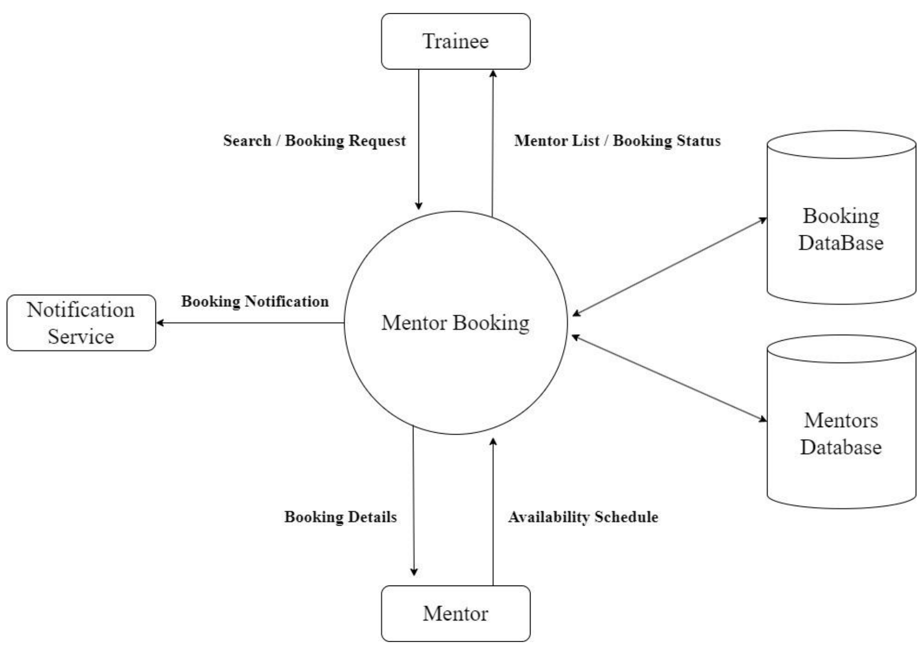


Figure 4.4: Data Flow Diagram – Process 3.0 Mentor Booking

Process 4.0 – Exam Management: Manages theory, medical, and practical exam booking and synchronization with the simulated government system.

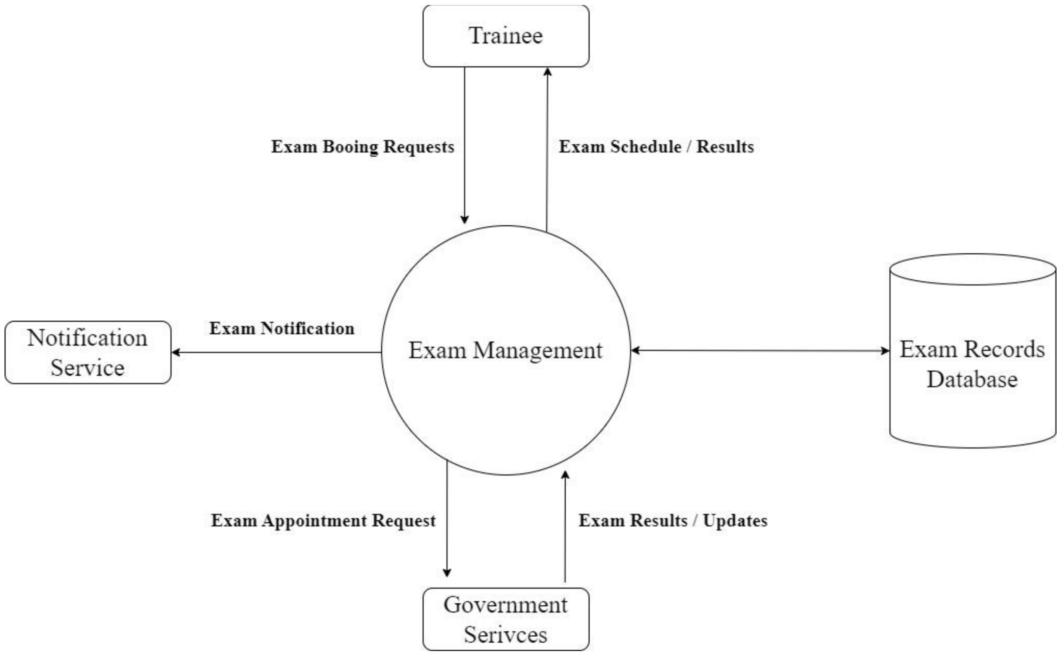


Figure 4.5: Data Flow Diagram – Process 4.0 Exam Management

Process 5.0 – Messaging & Notifications: Handles communication between trainees and mentors, notifications, and session feedback.

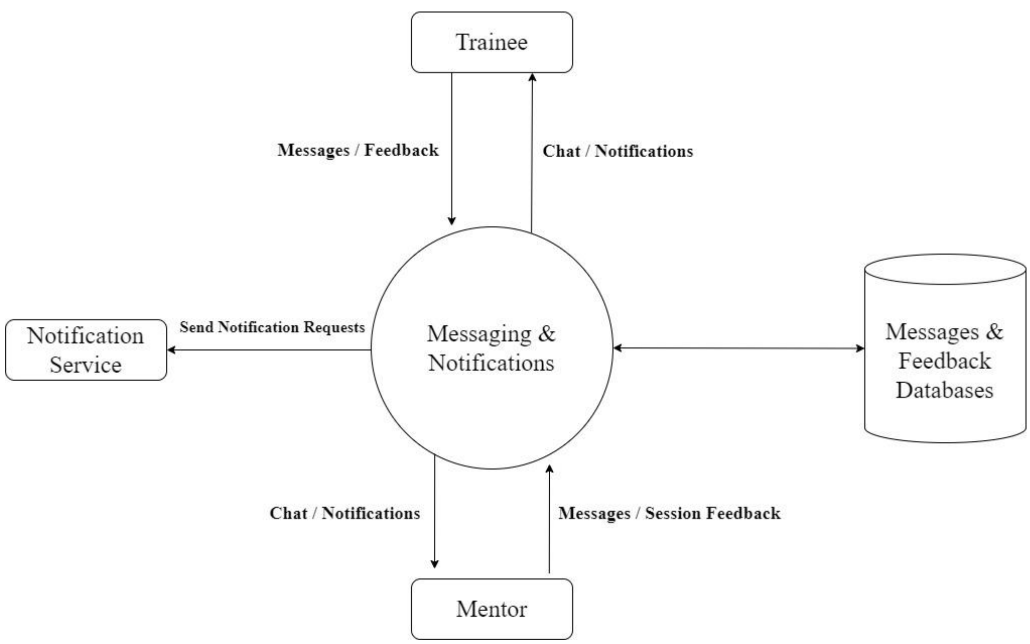


Figure 4.6: Data Flow Diagram – Process 5.0 Messaging and Notifications

Process 6.0 – Administration: Covers administrative functionalities such as mentor approval, analytics, content management, and monitoring logs.

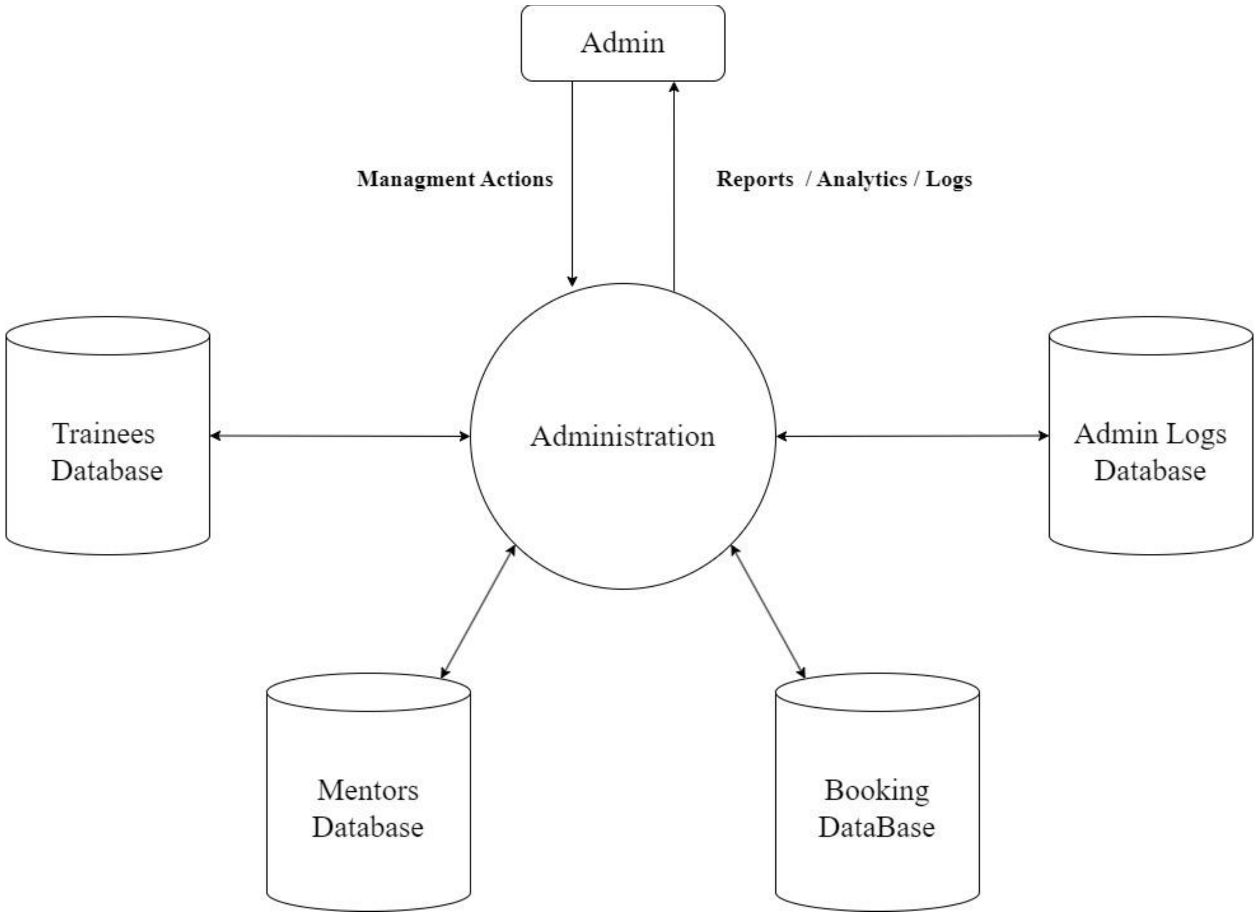


Figure 4.7: Data Flow Diagram – Process 6.0 Administration

This decomposition improves readability and allows each major subsystem to be represented clearly without overcrowding the diagrams.

4.4 Use Case Diagram

The Use Case Diagram illustrates the interactions between the different actors and the Ro5stak platform. It identifies the main functionalities provided by the system and demonstrates how trainees, mentors, administrators, and external services interact with these functionalities.

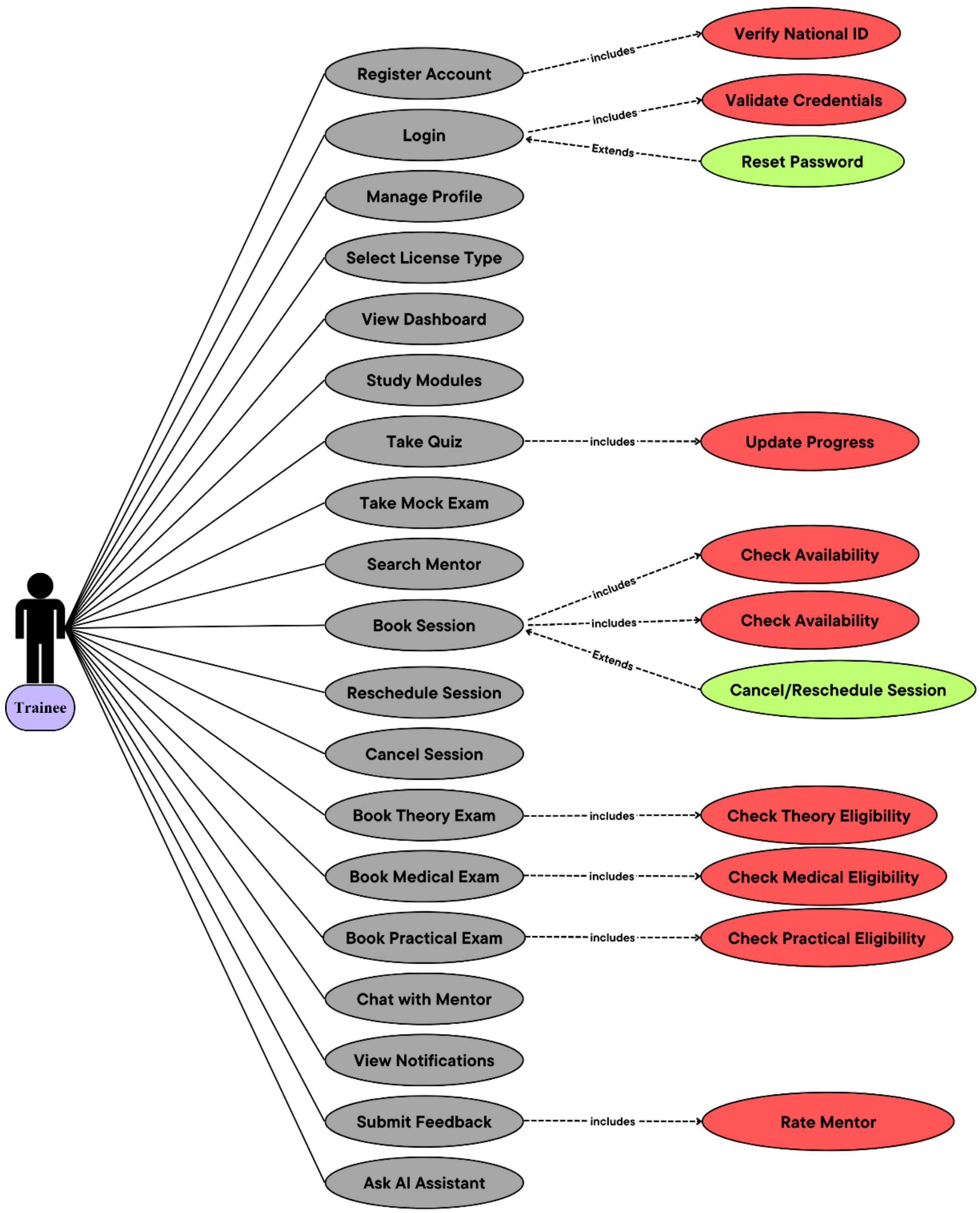


Figure 4.8: Use Case Diagram - Trainee

Figure 4.8: Use Case Diagram – Mentor And Admin

The diagram provides a high-level functional overview of the platform by showing the main use cases related to registration, learning, mentor booking, exam management, messaging, notifications, and administration. It also highlights the relationships between use cases using include and extend relationships where applicable.

4.5 Sequence diagram

Sequence Diagrams are used to represent the interaction flow between system components over time for specific scenarios. Since the Ro5stak platform contains a large number of functionalities and interactions, it would be impractical to model every possible sequence within the system. Therefore, only the most critical and representative workflows were selected.

The implemented sequence diagrams focus on the following key scenarios:

- Trainee Registration Process
- Mentor Booking Process
- Quiz and Progress Tracking Process

Trainee Registration Process :

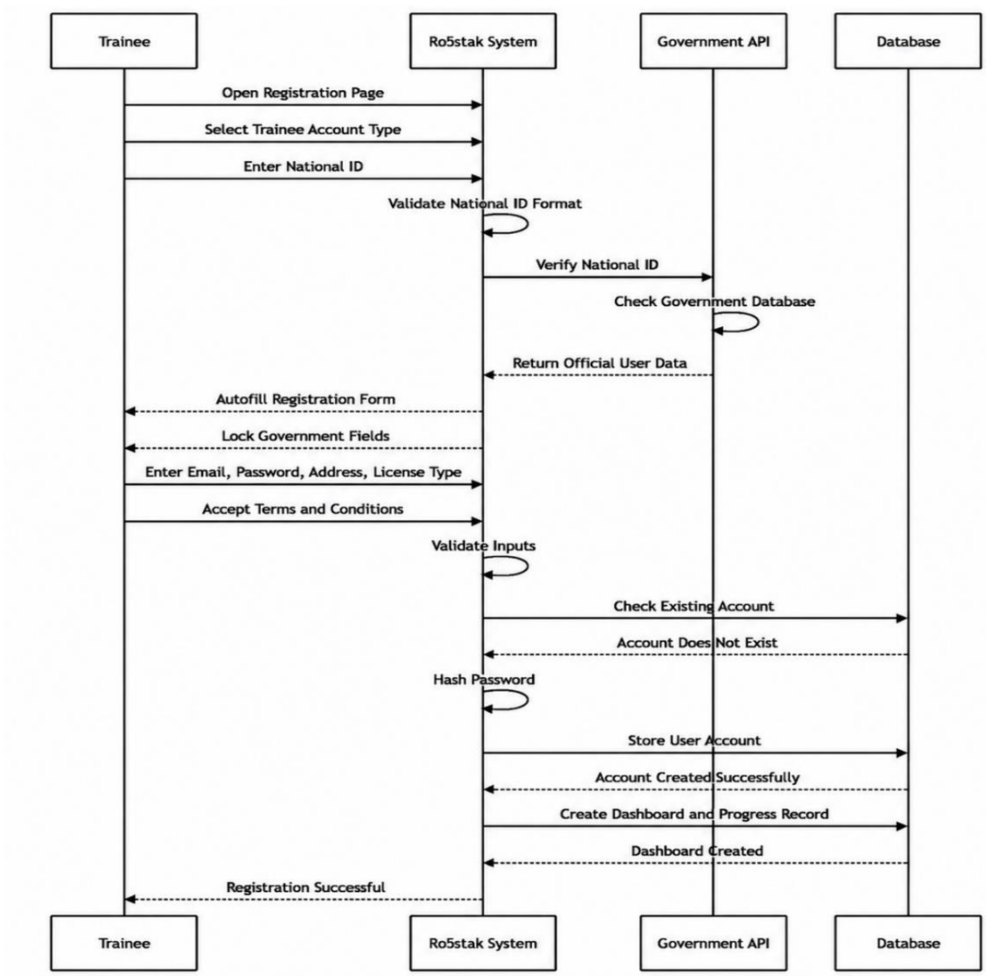


Figure 4.9: Sequence Diagram - Trainee Registration Process

Mentor Booking Process:

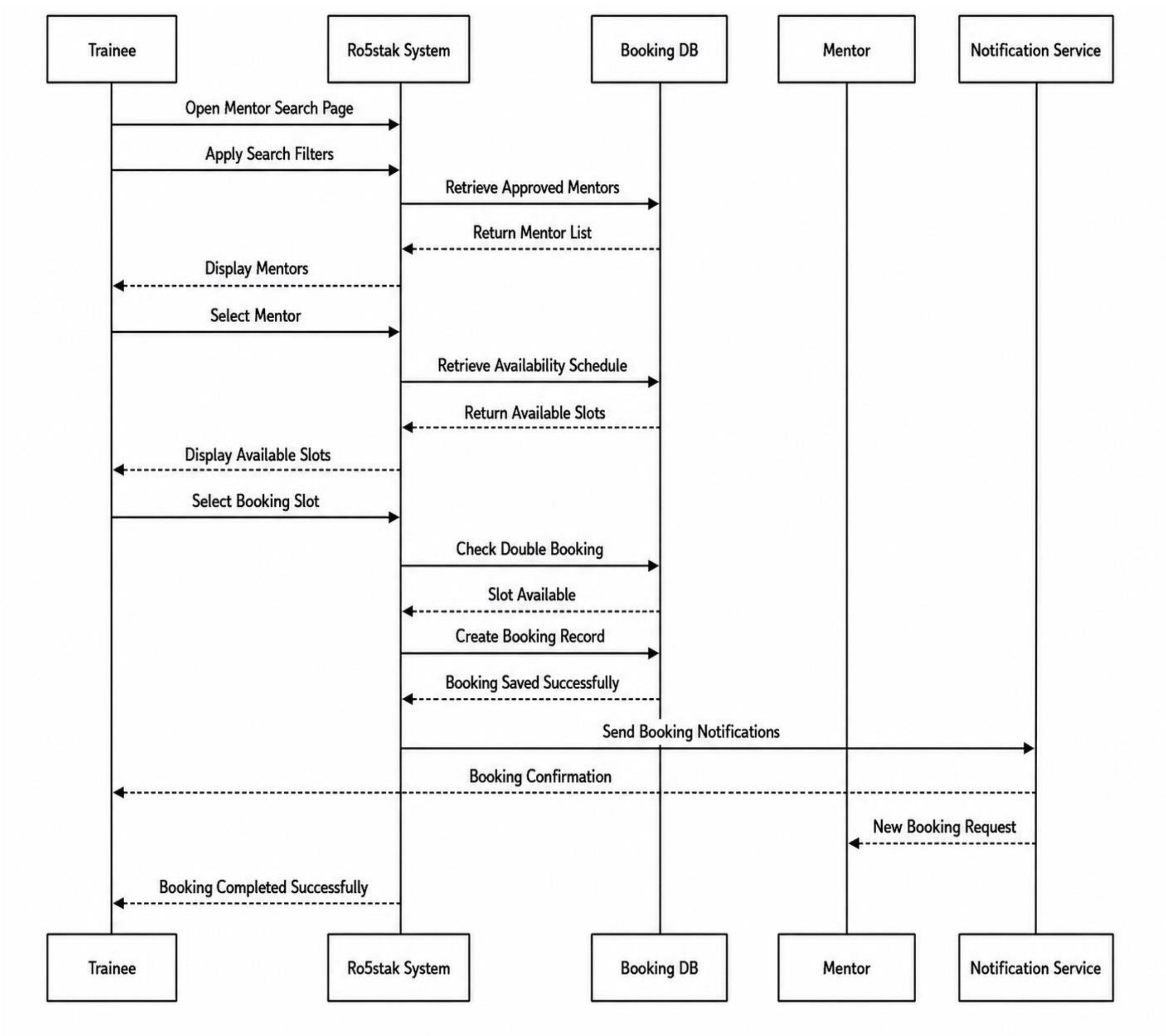


Figure 4.10: Sequence Diagram – Mentor Booking Process

Quiz and Progress Tracking Process:

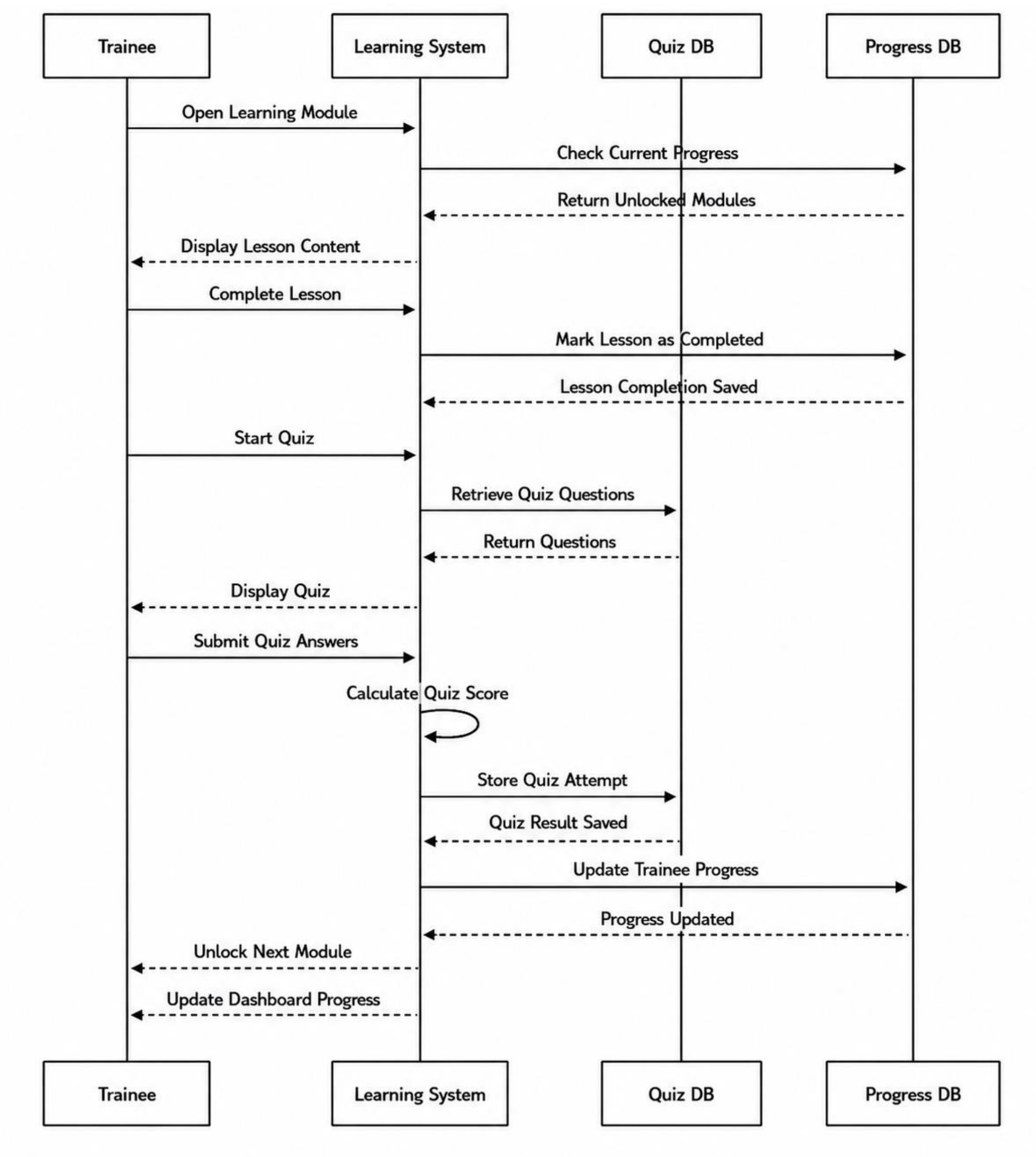


Figure 4.11: Sequence Diagram - Quiz and Progress Tracking Process

These scenarios were selected because they represent the core functionality of the platform and demonstrate important interactions between users, system services, databases, and external systems.

4.6 Activity diagram

Activity Diagrams were used to model the workflow and behavioral flow of major processes within the system. Due to the complexity of the platform and the large number of possible activities, only the most important workflows were selected for representation.

The implemented activity diagrams include:

Licensing Journey Workflow: representing the complete trainee journey from registration until completing the licensing process.

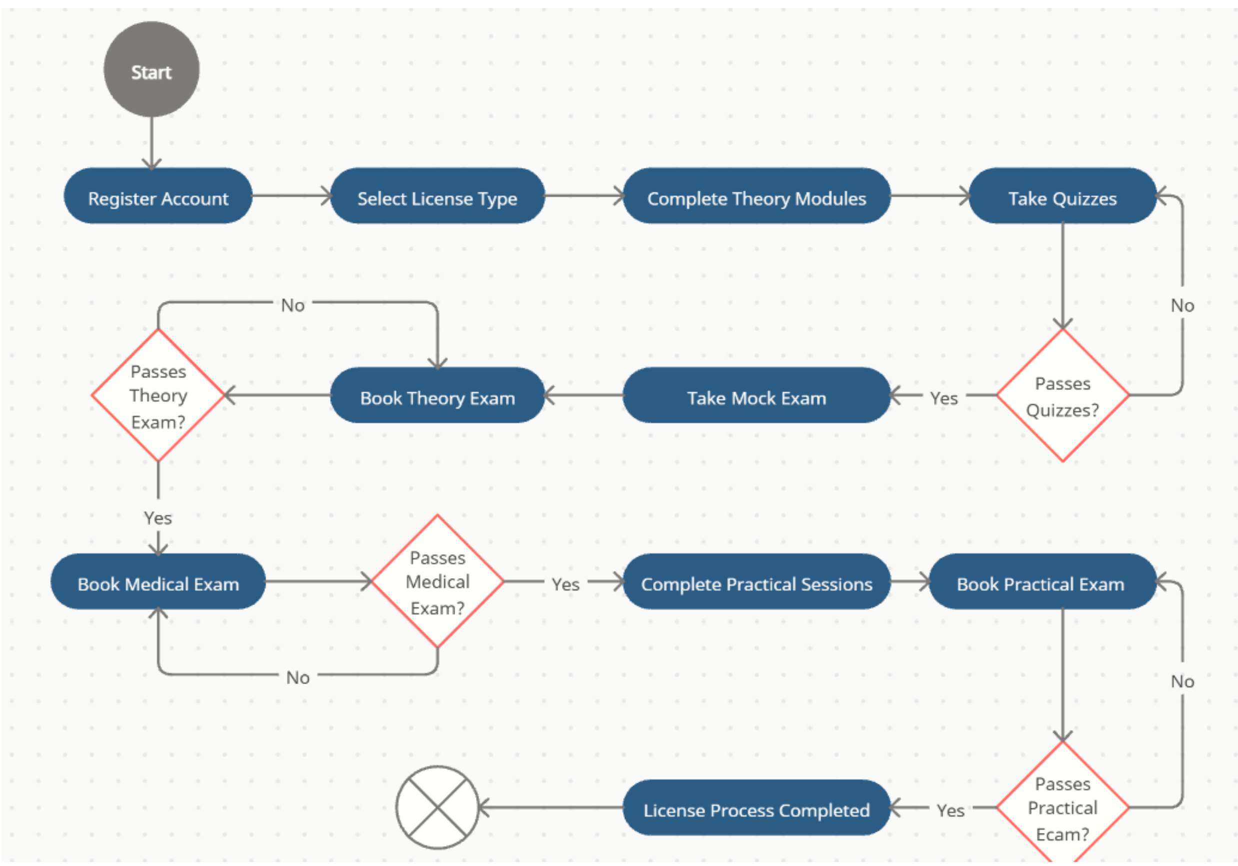


Figure 4.12: Activity Diagram - Licensing Journey Workflow

Mentor Approval Process: representing the workflow of mentor registration, document submission, administrative review, and approval or rejection.

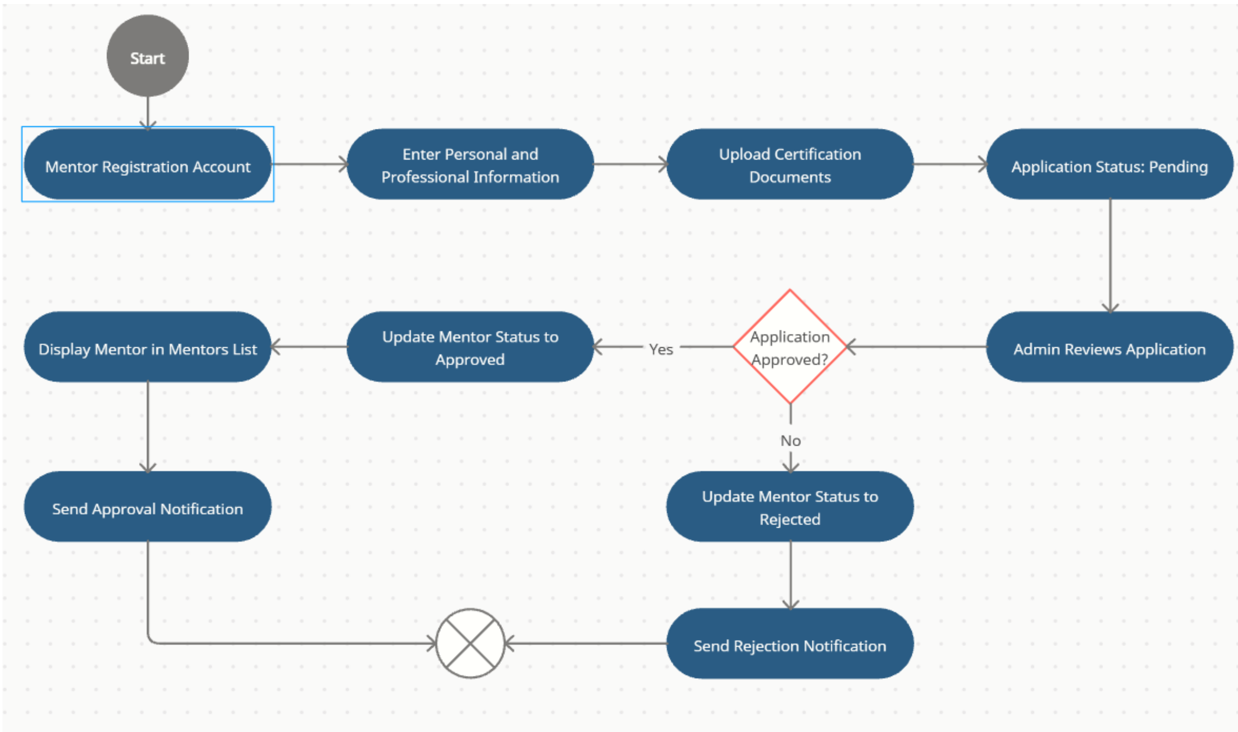


Figure 4.13: Activity Diagram - Mentor Approval Process

These workflows were selected because they represent critical operational processes within the platform and provide a clear understanding of the system's behavior and decision-making flow.

4.7 ER Diagram

The Entity Relationship Diagram (ERD) represents the core conceptual database structure of the Ro5stak platform before the normalization process. It illustrates the main entities within the system, their attributes, and the relationships between them, providing a foundational overview of the system's data organization.

The presented ERD focuses on the core concepts of the project and serves as the initial database design model used during system analysis and design. After creating the conceptual design, the database structure was further refined through the normalization process in order to reduce redundancy, improve data consistency, and ensure efficient database organization.

Several normalization examples and transformations will be presented later in this chapter to demonstrate how the database schema was improved and optimized during the design phase.

Chapter 5: Experiments and Results

5.1 Overview

5.2 Testing methodologies

5.2.1 Unit Testing Results

5.2.2 Integration Testing Results

5.2.3 System Testing Results

5.2.4 Acceptance System Results

5.3 Discussion and evaluation

5.4 Evaluation Metrics

5.5 Model Performance Comparison

5.6 Model Training Process

Chapter 6: Conclusion and Future Works

6.1 Overview

This chapter presents the final conclusions and outcomes of the proposed system. It summarizes the project idea, highlights the achieved objectives, discusses the main contributions and limitations, and presents possible future improvements. The chapter reflects the overall development journey of the system, starting from requirements gathering and system analysis to implementation and evaluation. It also demonstrates how the project addresses real-world problems and provides a practical and scalable solution for learners and mentors within the driving education domain.

6.2 Summary about the project

The project aims to develop an intelligent web-based platform that connects trainees with certified mentors to improve the driving learning experience through modern digital solutions. The system provides features such as mentor booking, online communication, learning modules, license management, official exam preparation, and personalized educational services within a centralized environment.

The platform was designed using modern software engineering methodologies and technologies to ensure scalability, maintainability, and usability. Different system models and diagrams were developed during the analysis and design phases, including ER diagrams, DFDs, use case diagrams, sequence diagrams, and activity diagrams. The implementation process focused on creating an organized and user-friendly system that simplifies communication between trainees and mentors while enhancing the learning experience through structured educational services.

6.3 Achieved objectives

The project successfully achieved the main objectives that were identified during the planning and requirements engineering phases. A complete web-based system was designed to support trainees, mentors, and administrators through different functionalities and services. The platform provides secure authentication and authorization mechanisms to manage different user roles efficiently.

Additionally, the system enables trainees to browse mentors, book sessions, communicate through integrated messaging, access learning materials, and manage their driving license journey digitally. The project also achieved a structured database design with normalized relations, responsive interfaces, and scalable architecture principles that support future enhancements and additional services.

6.4 Main contributions of the work

One of the main contributions of this project is providing a centralized digital platform that combines multiple driving education services within a single integrated system. Instead of relying on traditional communication and manual management methods, the platform automates major processes such as mentor applications, bookings, communication, and educational content management.

Another important contribution is the system architecture and database design, which were carefully planned to ensure scalability, maintainability, and flexibility for future development. The project also contributes academically by applying software engineering concepts such as requirements engineering, system modeling, database normalization, layered architecture, and modern web development technologies in a real-world case study.

6.5 Limitation

Despite the successful implementation of the project, there are several limitations that affected the scope of the current version. Due to time constraints and the nature of the project as a graduation project prototype, some advanced features were simplified or postponed for future development. For example, the notification system was limited mainly to email notifications, and some advanced AI-based functionalities were not fully integrated in the current release.

In addition, the platform was developed and tested within a limited environment and user base, which means large-scale performance and real-world deployment scenarios were not fully evaluated. Some external integrations and mobile application support were also outside the scope of the current implementation.

6.6 Future Work

Several improvements and additional features can be implemented in future versions of the project to enhance the system capabilities and user experience. One possible enhancement is developing a dedicated mobile application to provide easier accessibility for trainees and mentors. Advanced notification systems such as SMS and push notifications can also be integrated to improve communication efficiency.

Future versions may also include AI-powered recommendation systems to suggest mentors, personalized learning plans, and intelligent performance analysis for trainees. Additional security enhancements, payment gateway integration, live session support, analytics dashboards, and multilingual support can further improve the scalability and professionalism of the platform.

6.7 Conclusion

In conclusion, the project successfully demonstrates the development of an intelligent and integrated platform for driving education and mentorship management. The system provides a modern solution that improves communication, organization, and accessibility within the learning process while applying important software engineering principles and development methodologies.

The project achieved its primary objectives and delivered a functional prototype capable of supporting trainees and mentors through multiple educational and management services. Although there are some limitations, the platform establishes a strong foundation for future development and expansion, making it a promising solution within the digital learning and driving education sector.

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Appendices